

PVCA: Privacy-Preserving and Verifiable Cross-System Authorization for Platoon Communications in VANETs

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Abstract— Vehicle platoon, an increasingly significant technology in vehicular ad hoc networks (VANETs), effectively reduces energy consumption and environmental pollution, mitigates traffic congestion, and enhances road capacity and traffic safety. Collaborative communication between platoons is available to promote the reliability of data dissemination, ameliorate driving strategies, and further strengthen traffic efficiency. Nevertheless, existing efforts for secure multi-platoon data dissemination still face the following issues: (i) lack of a direct authorization strategy for platoons working on different systems; (ii) no effective mechanism to protect vehicle privacy during platoon communications; (iii) absence of a practical verification approach for cross-system ciphertext transformation. This paper builds a privacy-preserving and verifiable cross-system authorization (PVCA) scheme to mitigate the above issues. Specifically, we first put forth the optimized anonymous identity-based broadcast encryption (IBBE) and policy-hiding attribute-based encryption (ABE) protocols to adapt platoon communications while protecting identity and attribute privacy. After that, the authorized token bridging the proposed protocols is designed to enable ciphertexts of IBBE format to be transformed into new ciphertexts of ABE format, enabling flexible authorization. Furthermore, inspired by the Fujisaki-Okamoto transformation, we devise an efficient zero-knowledge proof of knowledge protocol, making our PVCA achieve verifiability and fairness. Rigorous security proof and analysis demonstrate that our PVCA is not only secure against chosen plaintext attacks and collusion attacks, but also satisfies the necessary security requirements. We implement a prototype of PVCA (on a desktop computer and Raspberry Pi) and provide a simulation utilizing NS-2 to validate its feasibility for platoon communications in VANETs. The results indicate that, compared to the state-of-the-art ciphertext transformation solutions, PVCA reduces the running time for original ciphertext generation, transformation, and decryption by at least 91.57%, 49.87%, and 50%, respectively, while compressing the original ciphertext and authorization token sizes by over 78.07% and 15.20%.

Index Terms—Cross-system authorization, fairness, privacy preserving, VANETs, verifiability.

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I. INTRODUCTION

VEHICULAR ad hoc networks (VANETs), as the essential part of the intelligent transportation system, connect intelligent vehicles via wireless links for real-time communication between other connected vehicles or nearby roadside infrastructures, effectively improving road safety and driving experience [1], [2]. In VANETs, organizing multiple vehicles into a platoon offers a promising solution to the problems of carbon emissions and traffic congestion caused by the proliferation of vehicles [3]. Typically, a platoon is made up of a platoon header (PH) vehicle and several platoon member (PM) vehicles. Through vehicle-to-vehicle (V2V) communication, the normal traveling PH is responsible for disseminating a series of command data to the PMs to ensure the proper speed and smaller vehicle spacing. Due to decreased inter-vehicle distances, vehicle platoon not only improves road capacity, but also reduces air resistance, further diminishing fuel consumption and environmental pollution [4], [5].

While vehicle platoon has attracted widespread attention and research benefiting from its various merits, there are also security and privacy concerns raised by the data transmitted over public communication channels in VANETs [6]. The disseminated data comprises numerous sensitive information about vehicles, for example, location, route planning, destination, car number, identity, etc. Criminal acts may be committed by a malicious adversary exploiting these information to deduce aspects of the driver's interests, beliefs, habits, and home address [7]. Encryption-then-dissemination paradigm provides an effective approach to sensitive information leakage problems based on the cryptographic primitives, e.g., identity/attribute-based encryption (IBE/ABE) [8], [9] and identity-based broadcast encryption (IBBE) [10]. Specifically, the PH utilizes the public keys of PMs to encrypt the data and then disseminates ciphertexts, ensuring the data is accessible only to authorized entities.

In vehicle platoon, merging two platoons with the same destination or adjacent to each other facilitates the improvement of traffic efficiency and reliability of data dissemination. Assuming that the two platoons (P and Q) are traveling on a road in an identical direction and platoon Q needs to merge into platoon P at some moment. Namely, the vehicles of platoon Q should have access to the disseminated data of platoon P . To guarantee confidentiality, the PH of P (i.e., PH^P) encrypts the platoon data with the public key of PH of Q (i.e., PH^Q). Upon receiving the

ciphertext, PH^Q needs to decrypt-then-encrypt the ciphertext and disseminate new ciphertexts for PMs of Q , which obtain the data and execute the commands by decrypting the ciphertext. Considering the delay in communication, the above process will undoubtedly disrupt the real-time nature of the commands, or worse, cause traffic accidents. In addition, the cryptosystems employed may vary between platoons owing to differences in communication service providers or vehicle manufacturers. Cross-system ciphertext transformation [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21] is a viable solution to the above problems. Specifically, inspired by the re-encryption function of proxy re-encryption (PRE) [22], an authorization token can be issued to the roadside unit (RSU) for transforming one encryption format into another. However, there remain three issues to be tackled when applying them to platoon communications.

(i) *There is lacking a direct authorization strategy for platoons working on different systems:* For platoon communications with multiple PMs as receivers, it is more appropriate to employ cryptosystems supporting multi-receiver scenarios. However, existing schemes [14], [15], [16], [17], [18], [19], [20], [21] are primarily centered on transforming the single receiver ciphertext (e.g., IBE ciphertext) into multiple receiver ciphertext (e.g., ABE or IBBE ciphertext). It implies that in addition to generating the ciphertexts for the PMs of P , the PH^P is required to produce IBE ciphertext decryptable by PH^Q and send it to the RSU. Meanwhile, the PH^Q also issues an authorization token to the RSU, which generates an ABE/IBBE ciphertext decryptable by the PMs of Q via the IBE ciphertext. This is obviously unacceptable for latency-sensitive command data. On the contrary, it is more applicable for platoon communications to directly transform the ciphertext decryptable by the vehicles in P (e.g., IBBE ciphertext) into the ciphertext decryptable by the vehicles of Q (e.g., ABE ciphertext). Thus, it remains challenging to design a direct authorization strategy for platoons working on different systems in VANETs.

(ii) *There is no effective mechanism to protect vehicle privacy during platoon communications:* The identity/attribute set of receivers generally has to be attached to IBBE/ABE ciphertext for assisting with the decryption operation in the schemes [14], [15], [16], [17], [18], [19], [20], [21]. It implies that vehicle privacy in platoons will be compromised since adversaries in the public communication channels are able to eavesdrop the ciphertexts. Further, the adversaries may infer the drivers' personal privacy and trace the route of the platoon, even triggering criminal behavior. Anonymous identity-based broadcast PRE schemes [23], [24], [25] and policy-hiding attribute-based PRE schemes [26], [27] seem to promise inspiration on this issue. Regrettably, the anonymity of [23], [24] can be broken by brute force attacks and the data confidentiality of [25] cannot be guaranteed. Meanwhile, the scheme [26] achieves access control through AND-gate access structure, which is less expressive. Although the scheme [27] overcomes this problem by adopting linear secret sharing scheme (LSSS), it cannot determine the correctness of the decryption result, due to the fact that the delegatee fails to know whether its attributes match the

hidden policy. Hence, devising an effective mechanism to protect vehicle privacy during platoon communications continues to be challenging.

(iii) *There is still the absence of a practical verification approach for cross-system ciphertext transformation:* Since the operation of the RSU responsible for performing the ciphertext transformation is non-transparent, it may return incorrect transformation results for saving its local resources. This means that it is unknown whether the RSU honestly utilizes the authorization token to transform the original ciphertext. That is, the vehicles in Q are unable to assure whether the transformation result returned by the RSU is valid. Also, the RSU should be safeguarded from malicious accusations of vehicles in Q when carrying out transformation operations honestly. This concern may be alleviated by verifiable and fair PRE [24], [27], [28], [29], [30], in which a re-encryption verification mechanism is introduced. Specifically, based on message lock encryption, an additional commitment to data and random number is embedded in the original ciphertext and opened during decryption, thus achieving verifiability and fairness. However, during the verification of the schemes [24], [27], [28], [30], the original ciphertext and authorization token are required to be transmitted to the verifier, incurring additional communication overhead and compromising the authorization token. Alternatively, if the proxy provides the wrong original ciphertext and authorization token, the verification result of schemes [24], [27], [28], [30] will be incorrect. Furthermore, each authorization token in [29] is generated for a specific ciphertext, implying that the delegator must obtain the original ciphertexts and deliver authorization tokens frequently. Thus, the solutions [24], [27], [28], [29], [30] are neither practical nor provide a complete verification approach for PRE. Simultaneously, these schemes are presented on a single cryptosystem with no applicability to the platoon communications scenarios. Accordingly, planning a practical verification approach for cross-system ciphertext transformation remains an unexplored field requiring urgent study.

Motivated by the above challenges, we present a privacy-preserving and verifiable cross-system authorization (PVCA) scheme for platoon communications. The main contributions are as follows.

- To guarantee privacy-preserving of the vehicle identity privacy and further improve the efficiency of IBBE system, we construct an optimized anonymous IBBE (AIBBE) protocol, in which a dummy identity is added to the identity set so that no adversary can infer the real identity information. Also, the polynomial technique is adopted to ensure the correctness of decryption while decreasing the communication cost. Meanwhile, an efficient policy-hiding ABE (PH-ABE) protocol capable of integrating with the optimized AIBBE is put forward, which not only protects attribute privacy in ABE system, but also supports a large universe, i.e., the number of attributes is not limited and specified during system initialization.
- With carefully designed authorization token combining the optimized AIBBE and PH-ABE protocols, we put forth

our PVCA scheme, where the authorized RSU is enabled to directly transform the IBBE ciphertext decryptable by the vehicles in $\mathbf{P}$ into the ABE ciphertext decryptable by the vehicles in $\mathbf{Q}$. More precisely, when platoons $\mathbf{P}$ and $\mathbf{Q}$ are required to be merged, $\text{PH}^{\mathbf{P}}$ issues the authorization token to the RSU, permitting the vehicles in $\mathbf{Q}$ to access the disseminated command data of $\mathbf{P}$. In this way, there is no need for $\text{PH}^{\mathbf{P}}$ to generate a special IBE ciphertext for $\text{PH}^{\mathbf{Q}}$, greatly reducing the computation and communication burden. In addition, even if the RSU colludes with the vehicles in $\mathbf{Q}$, our PVCA guarantees that the authorization token can not disclose the identity secret key of $\text{PH}^{\mathbf{P}}$.

- Taking inspiration from the Fujisaki-Okamoto transformation [31], our PVCA incorporates data into the random number so that the vehicles in $\mathbf{Q}$ can check the correctness of cross-system ciphertext transformation result by recovering the random number. Exploiting the devised zero-knowledge proof of knowledge protocol, the delegatee can produce publicly verifiable proofs to accuse the malicious behavior of RSU without revealing any privacy information. In addition, it is able to discourage malicious accusations against the RSU.
- We rigorously prove PVCA not only provides resistance to chosen plaintext and collusion attacks but also realizes verifiability and fairness. The performance evaluation demonstrates that our scheme is more viable and practical for platoon communications than related schemes.

The remainder of this paper is structured as follows. Section II reviews the related work. The preliminaries and system architecture are introduced in Sections III and IV, respectively. Section V describes the proposed PVCA in detail. Section VI presents the performance evaluation between our PVCA and other schemes. Section VII concludes this paper.

II. RELATED WORK

A. Proxy Re-Encryption and Cross-System Ciphertext Transformation

In 1998, Blaze et al. [22] suggested the conception of PRE, where a proxy is permitted to transform the ciphertext decryptable by the delegator into the ciphertext decryptable by the delegatee while not revealing the plaintext and secret key, achieving secure authorization. Considering the complex certificate management problems, Green et al. [32] combined IBE and PRE to develop the concept of identity-based PRE (IB-PRE) with the constructions against chosen plaintext/ciphertext attack (CPA/CCA). For achieving forward security and selective sharing, Liu et al. [33] proposed puncturable identity-based fine-grained proxy re-encryption and introduced a judge mechanism to check the behavior of the proxy. Zhang et al. [34] put forward a regulatable blockchain rewriting scheme, which combines IB-PRE and chameleon hash to achieve identity update. Aiming at supporting the sharing revoke in pub/sub-based systems, Li et al. [35] constructed a proxy-aided conditional IB-PRE scheme.

Regrettably, the above schemes only enable one-to-one sharing, hardly applicable to multi-receiver scenarios. Xu et al. [36]

integrated IBBE with PRE to propose the notion of identity-based broadcast PRE (IB-BPRE) with a concrete scheme that achieves constant-size ciphertexts. To support the revocation of delegatee in authorization token, Ge et al. [37] developed revocable IB-BPRE from the idea of revocable IBBE. However, the identity set is appended to the ciphertext in [36], [37], resulting in identity privacy leakage. Thus, Maiti et al. [23] presented the anonymous IB-BPRE scheme based on Lagrange interpolation polynomial technology. But [23] ignores the anonymity of original ciphertexts, i.e., the adversary can capture the identity information by brute force attack. Considering the unfixed ciphertext length in [23], Zhang et al. [25] designed a new anonymous IB-BPRE protocol. Nevertheless, the identities embedded in ciphertexts of the scheme [25] cannot contribute to achieving access control since any user with a valid secret key can obtain the plaintext from any ciphertext. With the possibility that the proxy may dishonestly transform ciphertexts, Sun et al. [24] investigated privacy-preserving, verifiable and fair IB-BPRE scheme based on inner product encryption. Unfortunately, an adversary could detect which identity vector matches the access vector in original and transformed ciphertexts, meaning that [24] fails to achieve identity privacy protection. An IB-BPRE scheme with anonymity and revocation [38] is put forth by combining [36] and the revocation idea in [37]. However, it suffers from heavy computation and communication costs. Chen et al. [39] tried to utilize the idea of blind signature to guarantee the verifiability of ciphertexts in IB-BPRE, i.e., not being replaced or tampered with. Whereas, in [39], the master key is compromised by any registered user, and the attackers can replace the ciphertext without being discovered.

For providing fine-grained access control, Liang et al. [40] constructed the attribute-based PRE (AB-PRE) scheme through empowering ABE with delegation capabilities. Zhang et al. [26] established the policy-hiding AB-PRE scheme to safeguard attribute privacy by appending a test component to the ciphertext. However, the scheme [26] achieves access control through AND-gate access structure, which is less expressive. To further improve the expressiveness of access policy, Zhang et al. [41] leveraged LSSS to put forth a fine-grained PRE scheme for cross-attribute domains, while the underlying plaintext is divulged to the proxy due to the insecure authorization token design. Considering the key leakage risk, Zhao et al. [42] put forward time-based AB-PRE with decryption key update, ensuring that the compromise of a decryption key for a certain time will not affect the security of other ciphertexts. Duan et al. [43] constructed the multi-authority AB-PRE (MAB-PRE) scheme to eliminate the centralization concerns. In order to prevent the data from being shared with malicious data receivers, Zhong et al. [44] designed an authorization token sanitization mechanism. However, in [44], the collusion of the edge server and any registered user will break the confidentiality of all plaintext. Aiming at the malicious behavior of untrustworthy proxy and delegates, Ge et al. [28] presented the verifiable and fair AB-PRE (VF-AB-PRE) scheme based on message lock encryption. Subsequently, VF-AB-PRE schemes [27], [29] with support for attribute privacy protection and bilateral access control are proposed following the idea of [28].

However, the above schemes are constructed for the same cryptosystem, where the original and transformed ciphertexts are in the same format, rendering them impossible to be deployed in cross-system communication. Therefore, Matsuo et al. [11] first proposed the cross-system ciphertext transformation scheme to transform certificate-based public key encryption (PKE) format into IBE format. The scheme in [12] realizes the mutual transformations made by PKE and IBE formats. The cross-system ciphertext transformation systems [13], [14], [15], [16], [17] between IBE and IBBE formats have also been suggested to fulfill the requirements of multi-user scenarios. Meanwhile, by bridging IBE and ABE, scholars also investigated the cross-system ciphertext transformation schemes [18], [19], [20], [21], realizing fine-grained access control. Nevertheless, there is still a lack of ciphertext transformation strategy between IBBE and ABE, and the above schemes cannot provide anonymity and verifiability.

B. Secure Platoon Communication in VANETs

Currently, scholars have suggested some privacy-preserving schemes to facilitate secure platoon communication in VANETs. By encapsulating the platoon key in an ABE ciphertext, Jiang et al. [45] designed a platoon sensing method to defend against various malicious attacks. Targeting misbehaving vehicle eavesdropping and data tampering, Lai et al. [46] constructed a privacy-preserving platoon communication protocol, where group key agreement and certificateless ring signcryption techniques are employed for intra and inter platoon communication, respectively. Zhao et al. [47] built a revocable ABE mechanism to support effective vehicle leaving in platoon and employed edge computing units to minimize the decryption overhead of vehicles through outsourced decryption. However, the latency will be increased since the edge computing unit needs to perform multiple outsourced decryption processes for the same ciphertext using different outsourced keys corresponding to different PMs. To guarantee the authenticity of disseminated command data, broadcast signcryption and bilateral access control schemes [48], [49], [50] are introduced. Wang et al. [51] took into account the heterogeneity between the platoon header and platoon member and established a multi-message and multi-receiver heterogeneous signcryption (MMHSC) system to improve the communication efficiency, which employs smart contracts to manage the platoon. Nevertheless, the above schemes do not consider the situation of platoon merging. Sun et al. [16] constructed a cross-domain authorization scheme to deal with communication between different platoons, allowing the conversion of IBE ciphertext to IBBE ciphertext. Subsequently, Zhao et al. [17] also investigated how to transform an identity-based signcryption ciphertext to an identity-based broadcast signcryption ciphertext.

Another series of work aimed at selecting an appropriate and reliable PH vehicle for platoon by designing reputation-based privacy-preserving schemes. Zhang et al. [52] put forth a trust-based platoon recommendation protocol, which collects the feedback of PMs and utilizes a weighted method to update the reputation score of PH after finishing a trip with a platoon.

Meanwhile, it utilizes Paillier cryptosystem and pseudonyms to protect vehicles' privacy. By recording reputation values on blockchain to provide non-repudiation and tamper resistance, Li et al. [53] built a trust-aware and decentralized speed advisory system, where the incentive mechanism is devised to mitigate the potential disincentives of increased vehicle energy consumption. Through designing the secure comparison protocol, a reputation-based platoon management mechanism [54] is introduced to select the PH vehicle without revealing the reputation values of vehicles. In order to facilitate the formation of high-quality platoons, Xu et al. [55] merged vehicle attribute information and proposed a multi-dimensional privacy-preserving protocol, which adopts private set intersection to realize privacy-preserving attribute matching. Sun et al. [56] pointed out that existing schemes suffer from the single reputation attack and developed a privacy-preserving dual reputation management scheme, in which a score-difference calculation algorithm is adopted to enable the weighted averaging and dual reputation management mechanisms.

Table I provides a comprehensive analysis between related schemes and our PVCA, focusing on the underlying technology, advantages, and limitations. Different from previous proxy re-encryption and cross-system ciphertext transformation schemes, our PVCA enables direct transformation of IBBE ciphertext into ABE ciphertext. By constructing the optimized AIBBE and PH-ABE protocols, the vehicle privacy is thoroughly protected while optimizing efficiency. Meanwhile, in contrast to the commitment-based ciphertext transformation verification approach (which is incomplete as stated in Section I), PVCA combines the Fujisaki-Okamoto transformation with the devised zero-knowledge proof protocols, realizing complete and practical cross-system ciphertext transformation verification. It should also be noted that the linear growth in ciphertext size within our PVCA is an inherent characteristic of anonymous and secure broadcast communication scenarios. To minimize communication costs as much as possible, we employ polynomial technology, which constitutes one of the main design ideas for optimizing PVCA efficiency. Furthermore, as demonstrated in Section VI-C, the prototype evaluation confirms that PVCA achieves nearly optimal storage and communication costs compared to state-of-the-art schemes.

III. PRELIMINARIES

A. Bilinear Pairing and Security Assumption

Let $\mathbb{G}$ and $\mathbb{G}_T$ be both multiplicative cyclic groups with prime q order. For any $g, h \in \mathbb{G}$, and $\alpha, \beta \in \mathbb{Z}_q$, the bilinear pairing $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ satisfies the features: 1) $e(g^\alpha, h^\beta) = e(g, h)^{\alpha\beta}$; 2) $e(g, g) \neq 1_{\mathbb{G}_T}$; 3) It is efficient to calculate $e(g, h)$.

(P, Q) -General decisional Diffie-Hellman exponent ((P, Q)-GDDHE) assumption: Given the tuple $(g_0, g_0^a, \dots, g_0^{a^q}, g_0^{a \cdot P(a)}, h_0, h_0^a, h_0^{a^{2n}}, h_0^{s \cdot Q(a)}, Z)$, it is difficult to determine whether $Z = e(g_0, h_0)^{s \cdot P(a)}$ or is a random element in $\mathbb{G}_T$, in which $P(x) = \prod_{j=1}^q (x + \kappa_j)$ and $Q(x) = \prod_{j=q+1}^{q+n} (x + \kappa_j)$ are two polynomials with different roots $\kappa_1, \dots, \kappa_{q+n} \in \mathbb{Z}_q$.

TABLE I
COMPREHENSIVE SUMMARY OF RELATED SCHEMES

Scheme	Technology	Advantages	Limitations
[16]	PRE & IBME & IBBE	Provide security against collusion attacks; Constant ciphertext size.	High latency triggered by indirect authorization; Not consider identity privacy of vehicles.
[17]	PRE & IBSC & IBBS	Provide unlinkability of ciphertexts; Constant ciphertext size.	Not consider identity privacy of PMs; Fail to resist collusion attacks.
[19]–[21]	PRE & IBE & ABE	Achieve fine-grained access control; Provide collusion attack resistance.	Not consider attribute privacy; Ignore the correctness of transformation results.
[23]	PRE & IBE & IBBE	Provide anonymity of transformation ciphertexts.	Not consider anonymity of original ciphertexts; Ciphertext size with linear growth.
[24]	IB-BPRE & Commitment	Prevent from malicious accusation; Provide decryption result correctness check.	Incomplete verification mechanism; Suffer from brute force attacks.
[25]	IB-BPRE & Pseudonyms	Achieve constant decryption cost.	Fail to protect the data confidentiality; Rely on the tamper-proof device on vehicles.
[26]	AB-PRE	Realize fine-grained access control; Achieve anonymity of ciphertexts.	Less expressiveness; Fail to support a large universe.
[27]	AB-PRE	Enable expressive fine-grained access control; Support access policy partially hidden.	Fail to check whether the policy matches; Suffer from collusion attacks.
[28]	AB-PRE & Commitment	Prevent from malicious accusation; Provide decryption result correctness check.	Incomplete verification mechanism; Fail to support a large universe; Not consider attribute privacy.
[29]	AB-PRE & ME CHF & DS	Support verification of transformation ciphertexts; Provide verifiable outsourced decryption.	Impractical authorization token generation; High communication overhead; Fail to provide attribute privacy protection.
[33]	PIB-FPRE & IBS	Guarantee fine-grained forward security; Achieve fine-grained re-encryption.	Not applicable to multi-receiver scenarios; Not consider identity privacy of vehicle.
[36]	IB-BPRE	Realize constant-size ciphertexts; Support conditional authorization.	High decryption overhead; Not consider identity privacy.
[38]	IB-BPRE	Protect identity privacy; Support revocation from authorization token.	High computation and communication overhead; Ciphertext size with linear growth.
[41]	AB-PRE	Enable cross-attribute domains authorization.	Fail to protect the data confidentiality.
[43]	MAB-PRE	Decentralize the single authority.	High computation and communication overhead.
[44]	AB-PRE & ACE	Provide an authorization token sanitization mechanism.	Fail to support a large universe; Suffer from collusion attacks.
[45]	ABE	Support fine-grained access control; Provide security against various attacks.	Not consider attribute privacy of vehicles.
[46]	GKA & CLRS	Realize intra and inter platoon communication; Support tracking of disputed or misbehaving PMs.	High computation and communication overhead.
[47]	RABE	Enable direct revocation for vehicle leaving situations; Reduce vehicle decryption burden.	Not consider attribute privacy of PMs; High latency triggered by outsourced decryption.
[48]	IBBS	Guarantee authenticity of command data; Constant ciphertext size.	Not consider identity privacy of PMs; Fail to resist collusion attacks.
[49]	PH-ABE & ME	Provide anonymity for vehicles; Reduce vehicle encryption and decryption burden.	Less expressive access control; High latency triggered by outsourced decryption.
[50]	RABE & ABME	Preserve attribute privacy of PH; Enable direct revocation for vehicle leaving situations.	Not consider attribute privacy of PMs.
[51]	MMHSC & Blockchain	Achieve heterogeneous communication for platoons; Guarantee the data confidentiality and authenticity.	Not consider identity privacy of vehicle; High latency triggered by blockchain.
Ours	PRE & Optimized AIBBE & PH-ABE & ZKPOK	Support direct authorization for platoons; Protect identity/attribute privacy of vehicles; Provide complete ciphertext transformation verification.	Ciphertext size with linear growth.

ME: Matchmaking encryption; IBME: Identity-based ME; ABME: Attribute-based ME; IBSC: Identity-based signcryption; IBBS: Identity-based broadcast signcryption; CHF: Chameleon hash function; DS: Digital signature; PIB-FPRE: puncturable identity-based fine-grained proxy re-encryption; IBS: Identity-based signature; ACE: Access control encryption; GKA: Group key agreement; CLRS: Certificateless ring signcryption; RABE: Revocable ABE.

q -Parallel decisional bilinear Diffie-Hellman exponent (q -PDBDHE) assumption: Given the tuple $(g_0, g_0^s, g_0^a, \dots, g_0^{a^q}, g_0^{a^{q+2}}, \dots, g_0^{a^{2q}}, \{g_0^{a \cdot s \cdot b_k / b_j}, \dots, g_0^{a^q \cdot s \cdot b_k / b_j}\}_{k,j \in [1,q], k \neq j}, \{g_0^{s b_j}, g_0^{a/b_j}, \dots, g_0^{a^q/b_j}, g_0^{a^{q+2}/b_j}, \dots, g_0^{a^{2q}/b_j}\}_{j \in [1,q]}, Z)$, it is difficult to determine whether $Z = e(g_0, g_0)^{a^{q+1} \cdot s}$ or is a random element in G_T .

q -Strong Diffie-Hellman (q -SDH) assumption: Given the tuple $(g_0, g_0^a, \dots, g_0^{a^q})$, it is difficult to find the pair $(\kappa, g_0^{\frac{1}{\kappa+a}})$, in which $\kappa \in \mathbb{Z}_q$.

Collision-free hash (CFH) assumption: Given the cryptographic hash function $\mathcal{H}$, it is difficult to find the pair (m, m') satisfying $\mathcal{H}(m) = \mathcal{H}(m')$, in which $m \neq m'$.

B. Anonymous Identity-Based Broadcast Encryption (AIBBE)

AIBBE [57] permits a broadcaster to generate a broadcast ciphertext for a broadcast group without compromising the identity privacy of receivers. Only the receivers in this group can decrypt the ciphertext into the broadcast message. In order to

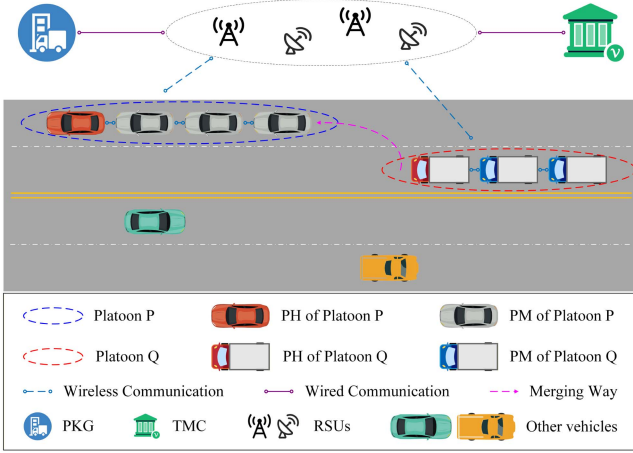


Fig. 1. System model of PVCA.

accommodate scenarios where multiple messages are required to transmit for a fixed group (e.g., vehicle platoon, military satellite communication, pay-TV, and so on), we introduce optimized AIBBE. Specifically, a group establishment algorithm is added to the definition of AIBBE [57] for pre-generating the broadcast group parameters. Formally, optimized AIBBE is defined as the following algorithms.

- $\text{Setup}_{\text{IBBE}}(\lambda)$. On inputting the security parameter λ , this algorithm outputs the public parameters PK_{IBBE} and master key MSK_{IBBE} .
- $\text{Extract}_{\text{IBBE}}(id, MSK_{\text{IBBE}})$. On inputting the identity id and master key MSK_{IBBE} , this algorithm outputs the IBBE secret key sk_{id} .
- $\text{Establish}_{\text{IBBE}}(S)$. On inputting the identity set S , this algorithm outputs the group parameters P_S .
- $\text{Encrypt}_{\text{IBBE}}(M, P_S)$. On inputting the data M and group parameters P_S , this algorithm outputs the IBBE ciphertext C_S .
- $\text{Decrypt}_{\text{IBBE}}(C_S, sk_{id_j})$. On inputting the IBBE ciphertext C_S and IBBE secret key sk_{id_j} , this algorithm outputs the data M' (if $id_j \in S$) or the symbol $\perp$.

More preliminaries, e.g., access structures, policy-hiding attribute-based encryption, zero-knowledge proof of knowledge, and digital signature, are provided in Appendix A, available online.

IV. SYSTEM ARCHITECTURE

This section gives the system architecture of PVCA, including the system model, formal definition, security requirements, and security model.

A. System Model

Fig. 1 demonstrates the system model of PVCA, consisting of private key generator (PKG), traffic manage center (TMC), roadside units (RSUs), and different platoons.

- **PKG**: The fully honest PKG has the main responsibility for setting up the system and issuing secret keys for vehicles.

- **TMC**: The TMC is the arbitration institution responsible for dealing with the accusations against RSU.
- **RSUs**: The untrusted RSUs act as the proxy to provide the cross-system ciphertext transformation service for platoons through wireless communication. Specifically, it utilizes the authorization token issued by the PH to transform the ciphertexts broadcast in a platoon into the ciphertexts accessible to the vehicles of another platoon.
- **Platoons**: Each platoon is composed of vehicles with the same destination, which maintain appropriate speeds and smaller spacing. Meanwhile, the vehicles in a platoon are categorized as platoon header (PH) and platoon members (PMs).
- **PH**: The trusted PH is in charge of establishing the platoon and periodically broadcasting command data to the PMs. Also, it is able to produce the authorization token to the RSU for sharing the data with another merged platoon.
- **PMs**: The semi-trusted PMs automatically follow the PH and adapt the speed in accordance with the PH's commands obtained from the decryption. Furthermore, PMs in the merged platoon could verify the correctness of transformation result and generate a proof to accuse the RSU's malicious behavior when the RSU indeed transformed the ciphertext incorrectly.

Note that in our PVCA, we assume that the platoon **P** works in the IBBE system, and the platoon **Q** working in the ABE system would like to merge into **P**. Meanwhile, there is at least one access policy matching the attributes of vehicles in platoon **Q**.

B. Formal Definition

By introducing the necessary algorithms for verifiable cross-system ciphertext transformation to bridge optimized AIBBE and PH-ABE, we give the formal definition of PVCA, which consists of the following algorithms.

- $\text{Setup}(\lambda) \rightarrow (MSK, PK)$. The setup algorithm receives the security parameter λ as input and outputs the master key MSK and public parameters PK .
- $\text{Extract}_{id}(id, MSK)$. The identity secret key extraction algorithm receives the identity id and master key MSK as inputs, and outputs the identity secret key sk_{id} .
- $\text{Extract}_{att}(A, MSK)$. The attribute secret key extraction algorithm receives the attribute set A and master key MSK_{ABE} as inputs, and outputs the attribute secret key sk_A .
- $\text{Establish}(S)$. The platoon establishment algorithm receives the identity set S as input, and outputs the platoon parameters P_S .
- $\text{Encrypt}(M, P_S)$. The encryption algorithm receives the data M and platoon parameters P_S as inputs, and outputs the original ciphertext C_S .
- $\text{Decrypt}_{or}(C_S, sk_{id_j})$. The original ciphertext decryption algorithm receives the original ciphertext C_S and secret key sk_{id_j} as inputs, and outputs the data M' if $id_j \in S$ or the symbol $\perp$.

- $\text{Authorize}(\mathbb{A}, id, sk_{id}, P_S)$. The authorization token generation algorithm receives the access policy $\mathbb{A} = (\mathbb{M}, \phi, \Gamma)$, identity id , secret key sk_{id} , and platoon parameters P_S as inputs, and outputs the authorization token AT , in which the attribute value set Γ is not contained in AT .
- $\text{Transform}(AT, C_S)$. The transform algorithm receives the authorization token AT and original ciphertext C_S as inputs, and outputs the transformed ciphertext C_A .
- $\text{Decrypt}_{tr}(C_A, sk_A)$. The transformed ciphertext decryption algorithm receives the transformed ciphertext C_A and secret key sk_A as inputs, and outputs false (when A fails to match $\mathbb{A}$), or $\perp$ (when C_A is incorrect), or the data M' .
- $\text{Claim}(C_A, sk_A)$. The claim algorithm receives the transformed ciphertext C_A and secret key sk_A as inputs, and outputs a proof π .
- $\text{Verify}(C_A, \pi)$. The verify algorithm receives the transformed ciphertext C_A and proof π as inputs, and outputs a bit 1 indicating that C_A is transformed incorrectly and 0 otherwise.

C. Security Requirements

PVCA should fulfill the following security requirements.

- *Platoon data confidentiality*: Only PH and authorized PMs have access to the disseminated command data, while others cannot acquire any information from the ciphertext. In addition, the vehicles in platoon $\mathbf{Q}$ with the attribute set matching the access policy have the capability to restore the command data from the transformed ciphertext, which is not readable by any adversary, including unauthorized vehicles and RSUs.
- *Identity and attribute privacy protection*: Anyone cannot learn the identities or attributes of PMs from the original/transformed ciphertext even if it is capable of decrypting the ciphertext. Furthermore, the authorization token held by the RSU does not compromise the identity/attribute privacy of PMs.
- *Verifiability and fairness*: The correctness of transform results can be verified by the vehicles in platoon $\mathbf{Q}$. Also, the honest RSU returning the correct results is free from malicious accusations.
- *Resistance to collusion attack*: Even if the PMs in platoon $\mathbf{Q}$ and RSU collude, they are neither able to deduce the identity secret key of PH in platoon $\mathbf{P}$, nor access unauthorized data.
- *Resistance to other attacks*: PVCA should have the ability to resist various classical attacks in VANETs, e.g., replay, modification, and impersonation attacks.

D. Security Model

The security model of PVCA includes indistinguishability against chosen plaintext attack at IBBE ciphertext (IND-CPA-IBBE) and ABE ciphertext (IND-CPA-ABE), verifiability, fairness, and leakage-resistance against collusion attack (LR-CA), as captured by the games of a probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ and challenger $\mathcal{C}$. The IND-CPA-IBBE/IND-CPA-ABE security ensures that, given two equal-length data, the

adversary cannot distinguish the corresponding challenge original/transformed ciphertexts. Verifiability and fairness provide the assurance that the adversary is not capable of producing either an incorrectly converted ciphertext to pass verification or a valid proof for a correct transformed ciphertext. In addition, the LR-CA security guarantees that the adversary fails to output a valid identity secret key even if it knows arbitrary attribute secret keys and authorization tokens. The detailed security definitions are demonstrated in Appendix B, available online.

V. PROPOSED PVCA SCHEME

A. Main Idea

Before introducing the detailed scheme, we present the main idea of our PVCA scheme, which aims to achieve direct authorization, identity and attribute privacy protection, as well as verifiable cross-system ciphertext transformation.

Direct authorization from IBBE to ABE: We borrow the idea of secure authorization approach [15] to achieve direct transformation from IBBE ciphertext format into ABE ciphertext format while ensuring nothing is leaked from PH^P 's secret key. More precisely, in authorization token, the identity secret key sk_{id} is blinded by the random group element $u^{t \cdot H_4(\chi_1)}$, where $H_4(\chi_1)$ is an ABE encapsulated key. When the RSU uses this authorization token to transform IBBE ciphertext, the transformed ciphertext will be the blinded data. Only those vehicles with an attribute set matching the access policy can restore $H_4(\chi_1)$ and then eliminate the blinding factor, recovering the data. By this, not only is direct authorization provided, but also the PH^P 's identity secret key is protected.

Identity privacy protection of platoon vehicles: We design an optimized AIBBE protocol (as demonstrated in Fig. 2) to protect identity privacy of PMs in original ciphertext and minimize the computation and communication costs. Specifically, we split the whole identity space into two parts: (i) normal identity space for realizing the identity secret key extract, encrypt, decrypt, and transform operations; (ii) dummy identity space utilized to guarantee identity anonymity of platoon vehicles. The hash function H_1 computes the hash value of normal identity id is $H_1(1||id)$, while that of dummy identity rid is calculated as $H_1(0||rid)$. Then, we embed a dummy identity rid in platoon parameters to provide the randomness such that the adversary cannot determine the vehicle identity set from the ciphertext by brute force attacks. Meanwhile, the polynomial $f(x)$ over $\mathbb{Z}_q$ is generated through the points $\{(H_3(w_k), \sigma) \in \mathbb{Z}_q^2\}_{k \in [1, l]}$, where σ is the IBBE encapsulated key and w_k is the secret information that can be restored by sk_{id_k} and c_1 . Thus, PMs could obtain σ by evaluating $\{H_3(w_k)\}_{k \in [1, l]}$ on $f(x)$. In this way, the transmission of entire identity set is avoided while ensuring the correctness of decryption and reducing the communication cost.

Attribute privacy protection of platoon vehicles: A novel PH-ABE protocol (as depicted in Fig. 3), which can be efficiently integrated with the optimized AIBBE protocol, is devised to prevent attribute information from being leaked in transformed ciphertexts, starting from the efficient ABE scheme [58]. The modifications of secret key and ciphertext structure allow our PH-ABE to support a large universe without affecting efficiency

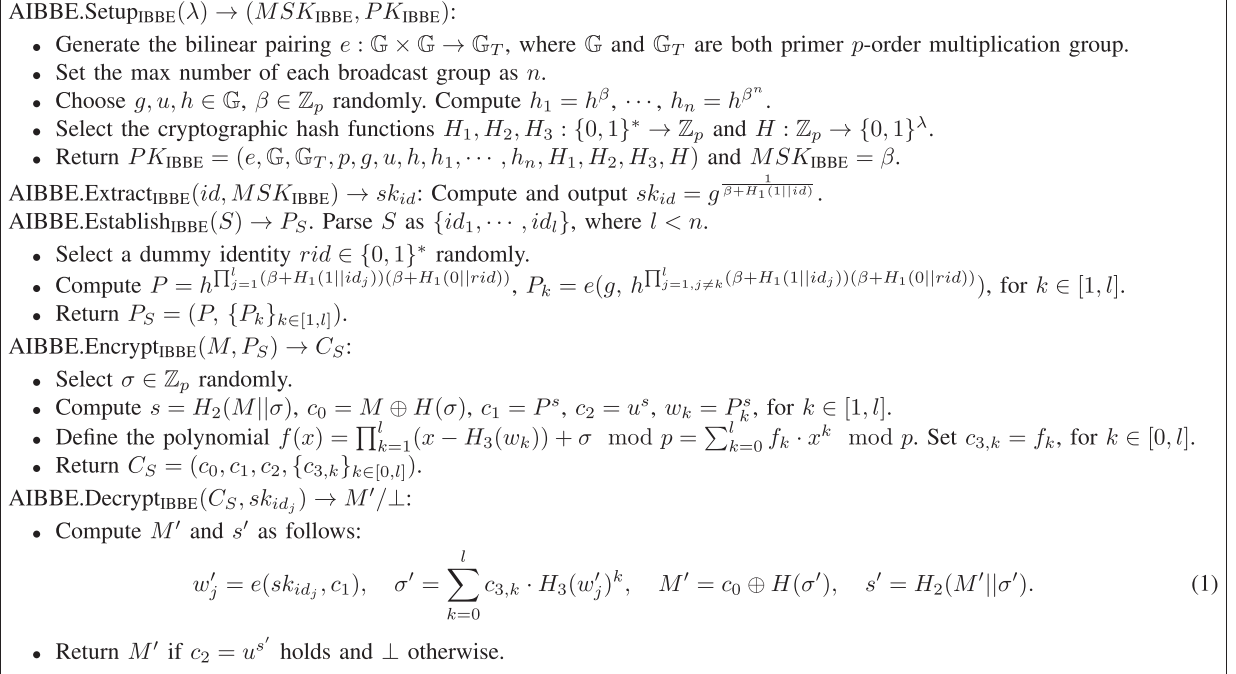


Fig. 2. Concrete construction of optimized AIBBE protocol.

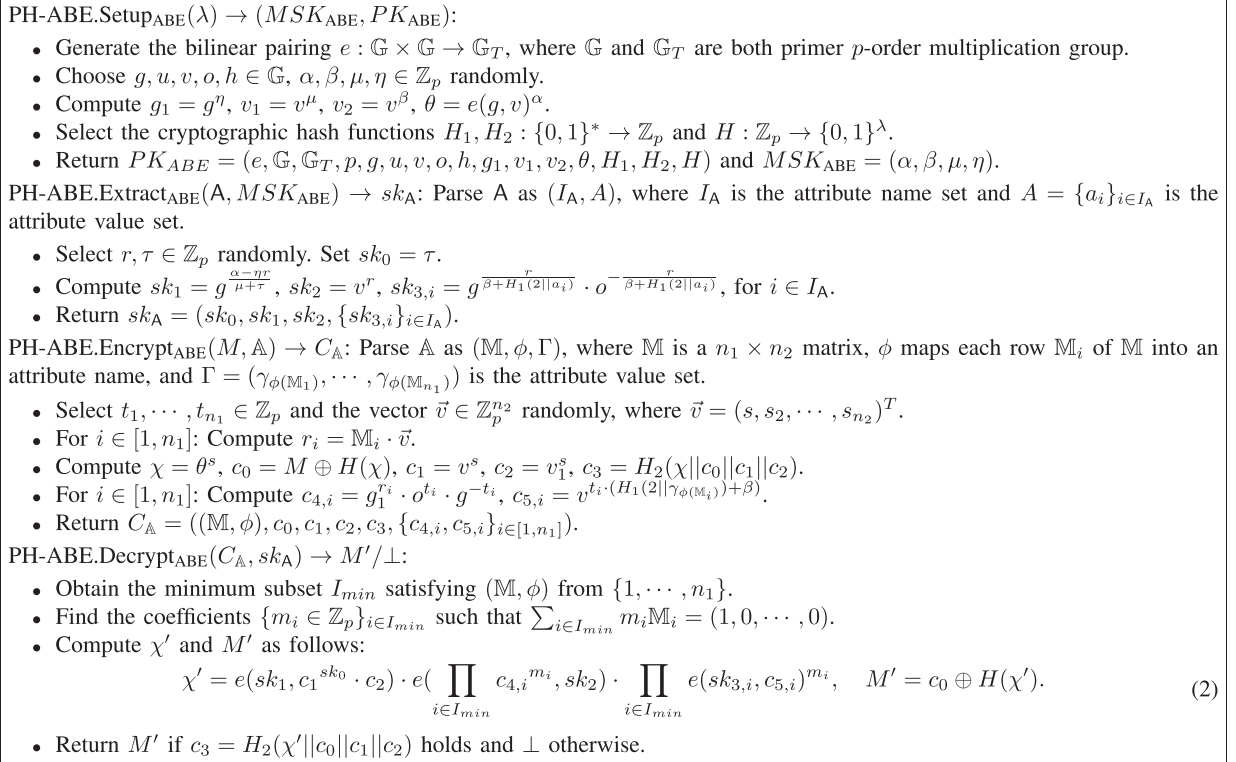


Fig. 3. Concrete construction of PH-ABE protocol.

$$\begin{aligned}
\Pi &= \{(sk_0, sk_1, sk_2, \{sk_{3,i}\}_{i \in I_{min}}) : \chi' = e(sk_1, c'_4)^{sk_0 \cdot c'_5} \cdot e(\prod_{i \in I_{min}} c'_{7,i}^{m_i}, sk_2) \cdot \prod_{i \in I_{min}} e(sk_{3,i}, c'_{8,i})^{m_i}\} \\
\text{Let } T_1 &= g^{t_1} \cdot h^{t_2}, T_2 = sk_1 \cdot g^{t_2}, \zeta_1 = sk_0 \cdot t_1, \zeta_2 = sk_0 \cdot t_2, \text{ it can be transformed to the standard } \Sigma\text{-protocol for the following} \\
\text{relation } \Pi', \text{ where } t_1, t_2 &\text{ are randomly selected from } \mathbb{Z}_p. \\
\Pi' &= \{(t_1, t_2, \zeta_1, \zeta_2, sk_0, sk_2, \{sk_{3,i}\}_{i \in I_{min}}) : T_1 = g^{t_1} \cdot h^{t_2} \wedge T_1^{sk_0} = g^{\zeta_1} \cdot h^{\zeta_2} \wedge \chi' / e(T_2, c'_5) = e(T_2, c'_4)^{sk_0} \cdot e(g, c'_4)^{-\zeta_2} \cdot \\
&\quad e(g, c'_5)^{-t_2} \cdot e(\prod_{i \in I_{min}} c'_{7,i}^{m_i}, sk_2) \cdot \prod_{i \in I_{min}} e(sk_{3,i}, c'_{8,i})^{m_i}\} \\
\text{ZKPoK.Proof}(sk_A, C_A): \\
&\bullet \text{ Randomly choose } t_1, t_2 \in \mathbb{Z}_p. \\
&\bullet \text{ Compute } T_1 = g^{t_1} \cdot h^{t_2}, T_2 = sk_1 \cdot g^{t_2}, \zeta_1 = sk_0 \cdot t_1, \zeta_2 = sk_0 \cdot t_2. \\
&\bullet \text{ Randomly choose } r_{t_1}, r_{t_2}, r_{\zeta_1}, r_{\zeta_2}, r_{sk_0} \in \mathbb{Z}_p, r_{sk_2} \in \mathbb{G}, r_{sk_{3,i}} \in \mathbb{G}, \text{ for } i \in I_{min}. \\
&\bullet \text{ Compute } R_1 = g^{r_{t_1}} \cdot h^{r_{t_2}}, R_2 = T_1^{-r_{sk_0}} \cdot g^{r_{\zeta_1}} \cdot h^{r_{\zeta_2}}, R_3 = e(T_2, c'_4)^{r_{sk_0}} \cdot e(g, c'_4)^{-r_{\zeta_2}} \cdot e(g, c'_5)^{-r_{t_2}} \cdot e(\prod_{i \in I_{min}} c'_{7,i}^{m_i}, r_{sk_2}) \cdot \\
&\quad \prod_{i \in I_{min}} e(r_{sk_{3,i}}, c'_{8,i})^{m_i}, c = H_5(\chi', T_1, T_2, R_1, R_2, R_3, C_A). \\
&\bullet \text{ Calculate } s_{t_1} = r_{t_1} - c \cdot t_1, s_{t_2} = r_{t_2} - c \cdot t_2, s_{\zeta_1} = r_{\zeta_1} - c \cdot \zeta_1, s_{\zeta_2} = r_{\zeta_2} - c \cdot \zeta_2, s_{sk_0} = r_{sk_0} - c \cdot sk_0, s_{sk_2} = r_{sk_2} \cdot sk_2^{-c}, \\
&\quad s_{sk_{3,i}} = r_{sk_{3,i}} \cdot sk_{3,i}^{-c}, \text{ for } i \in I_{min}. \\
&\bullet \text{ Outputs } \pi = \{\chi', \{m_i\}_{i \in I_{min}}, T_1, T_2, c, s_{t_1}, s_{t_2}, s_{\zeta_1}, s_{\zeta_2}, s_{sk_0}, s_{sk_2}, \{s_{sk_{3,i}}\}_{i \in I_{min}}\}. \\
\text{ZKPoK.Verify}(\pi, C_A): \\
&\bullet \text{ Calculate } R'_1, R'_2, \text{ and } R'_3 \text{ as follows:} \\
&\quad R'_1 = T_1^c \cdot g^{s_{t_1}} \cdot h^{s_{t_2}}, \quad R'_2 = T_1^{-s_{sk_0}} \cdot g^{s_{\zeta_1}} \cdot h^{s_{\zeta_2}}, \\
&\quad R'_3 = (\chi' / e(T_2, c'_5))^c \cdot e(T_2, c'_4)^{s_{sk_0}} \cdot e(g, c'_4)^{-s_{\zeta_2}} \cdot e(g, c'_5)^{-s_{t_2}} \cdot e(\prod_{i \in I_{min}} c'_{7,i}^{m_i}, s_{sk_2}) \cdot \prod_{i \in I_{min}} e(s_{sk_{3,i}}, c'_{8,i})^{m_i}. \quad (3) \\
&\bullet \text{ Return true if } c = H_5(\chi', T_1, T_2, R'_1, R'_2, R'_3, C_A) \text{ holds, and false otherwise.}
\end{aligned}$$

Fig. 4. Concrete construction of ZKPoK protocol.

and security. Furthermore, the access policy partially hidden method [59] is employed to achieve attribute anonymity of platoon vehicles, where the attribute names $(\phi(\mathbb{M}_1), \dots, \phi(\mathbb{M}_{n_1}))$ are publicly computed while corresponding attribute value set Γ is hidden in access policy $\mathbb{A}$. Meanwhile, an additional commitment is designed to allow the vehicles of platoon $\mathbf{Q}$ to check the correctness of the intermediate result. If correct, there is an indication that the attribute set matches $\mathbb{A}$, and the data can be further recovered.

Verifiability of cross-system ciphertext transformation: Inspired by Fujisaki-Okamoto transformation [31], the PH^P integrates the data M into the random number $s = H_2(M||\sigma)$ used to generate the IBBE ciphertext C_{SP} . By this means, the part c_2 of C_{SP} serves both as auxiliary information for cross-system transformation and the commitment for completing the verification of the ciphertext transformation process. Namely, after obtaining the data M' by decrypting the transformed ciphertext, the vehicles of platoon $\mathbf{Q}$ restore the random number s and verify the correctness of decryption result. If it passes, they accept M' , meaning that the RSU transforms the ciphertext honestly. Otherwise, they will accuse the malicious behavior of the RSU to the TMC. Specifically, through the designed ZKPoK protocol (as shown in Fig. 4) for the relation Π , the PM_j^Q proves that its attributes match $\mathbb{A}$ in the transformed ciphertext and the decryption auxiliary information χ'_1 is computed correctly. This approach ensures that the honest RSU is not subject to malicious accusations while guaranteeing there is no leakage of PM_j^Q 's secret key.

B. Detailed Construction of PVCA

By employing the optimized AIBBE ($\text{Setup}_{\text{IBBE}}$, $\text{Extract}_{\text{IBBE}}$, $\text{Enstablish}_{\text{IBBE}}$, $\text{Encrypt}_{\text{IBBE}}$, $\text{Decrypt}_{\text{IBBE}}$), PH-ABE ($\text{Setup}_{\text{ABE}}$, $\text{Extract}_{\text{ABE}}$, $\text{Encrypt}_{\text{ABE}}$, $\text{Decrypt}_{\text{ABE}}$), ZKPoK (Proof, Verify),

and DS (Setup_{DS} , $\text{KeyGen}_{\text{DS}}$, $\text{SigGen}_{\text{DS}}$, $\text{SigVerify}_{\text{DS}}$) protocols as the building blocks, we provide our PVCA scheme for platoon communications in detail, which comprises six phases, i.e., initialization, platoon establishment, platoon data dissemination, cross-platoon authorization, and cross-platoon access.

1) *Initialization:* In this phase, the PKG runs the algorithm $\text{Setup}(\lambda)$ to produce the master key MSK and issues the public parameters PK . Also, each vehicle registers with the PKG, which sends the identity/attribute secret key (i.e., sk_{id} or sk_A) to the vehicle by executing the algorithm $\text{Extract}_{id}(id, MSK)/\text{Extract}_{att}(A, MSK)$.

- *Setup:* Given the security parameter λ , this algorithm generates the bilinear pairing $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$, where $\mathbb{G}$ and $\mathbb{G}_T$ are both primer p -order multiplication group. The algorithm sets the max number of each IBBE platoon vehicle n , chooses $g, u, v, o, h \in \mathbb{G}$, $\alpha, \beta, \mu, \eta \in \mathbb{Z}_q$ randomly, and computes $g_1 = g^\alpha, h_1 = h^\beta, \dots, h_n = h^{\beta^n}, v_1 = v^\mu, v_2 = v^\beta, \theta = e(g, v)^\alpha$. The algorithm selects the cryptographic hash functions $H_1, H_2, H_3, H_4, H_5 : \{0, 1\}^* \rightarrow \mathbb{Z}_q$ and $H : \mathbb{Z}_q \rightarrow \{0, 1\}^\lambda$ and runs $\text{DS.Setup}_{\text{DS}}(\lambda) \rightarrow PK_{\text{DS}}$. Finally, it returns the master key $MSK = \{\alpha, \beta, \mu, \eta\}$ and public parameters $PK = (e, \mathbb{G}, \mathbb{G}_T, p, g, u, v, o, h, g_1, h_1, \dots, h_n, v_1, v_2, \theta, H_1, H_2, H_3, H_4, H_5, H, PK_{\text{DS}})$.
- *Extract_{id}:* Given the identity id and master key MSK , this algorithm returns the identity secret key sk_{id} by invoking the algorithm $\text{AIBBE.Extract}_{\text{IBBE}}(id, MSK)$.
- *Extract_{att}:* Given the attribute set $A = (I_A, A)$ and master key MSK , this algorithm returns the attribute secret key $sk_A = (sk_0, sk_1, sk_2, \{sk_{3,i}\}_{i \in I_A})$ by invoking the algorithm $\text{PH-ABE.Extract}_{\text{ABE}}(A, MSK)$.

2) *Platoon Establishment:* In this phase, the platoons $\mathbf{P}$ and $\mathbf{Q}$ under different systems are established. To guarantee the authenticity and integrity of the command data, the

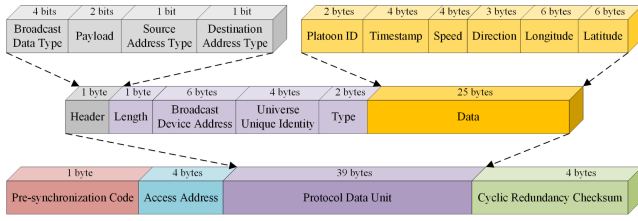


Fig. 5. Structure of the command data.

PH in each platoon produces the key pair $(sk_{sig}, vk_{sig}) = DS.KeyGen_{DS}(PK_{DS})$ and transmits the verification key vk_{sig} to the PMs. That is, the PH^P and PH^Q generate (sk_{sig}^P, vk_{sig}^P) and (sk_{sig}^Q, vk_{sig}^Q) , respectively. Also, through executing the algorithm $Establish(S_P)$, the PH^P with the identity id_1 pre-generates the parameters P_{S_P} of platoon P to reduce the computation cost during platoon data dissemination and stores P_{S_P} locally, in which $S_P = \{id_1, \dots, id_{l_P}\}$ is the vehicle identity set of platoon P .

- **Establish.** Given the identity set $S = \{id_1, \dots, id_l\}$ ($l < n$), this algorithm returns the platoon parameters $P_S = (P, \{P_k\}_{k \in [1, l]})$ by invoking the algorithm $AIBBE.Establish_{IBBE}(S)$.

3) *Platoon Data Dissemination:* In this phase, the PH^P wants to broadcast the command data m , which is a binary byte stream, and in frames. Fig. 5 illustrates the structure of command data [45]. The PH^P generates the signature $sig = DS.SigGen_{DS}(m, sk_{sig}^P)$, encrypts the data $M = m || sig$ to generate the original ciphertext C_{S_P} by running the algorithm $Encrypt(M, P_{S_P})$, and disseminates C_{S_P} to the RSU and PMs in platoon P . Upon receiving C_{S_P} , each PM_j^P restores $M' = m' || sig'$ through the algorithm $Decrypt_{or}(C_{S_P}, sk_{id_j})$, where $j \in [2, l]$. Then, if the command data m' is fresh (i.e., the difference between the timestamp in m' and current timestamp is less than the predefined threshold) and valid (i.e., $DS.SigVerify_{DS}(m, sig', vk_{sig}^P) = \text{true}$), the PMs take corresponding actions according to m' , e.g., adjusting the speeds.

- **Encrypt:** Given the data M , and platoon parameters P_S , this algorithm returns the original ciphertext $C_S = (c_0, c_1, c_2, \{c_{3,k}\}_{k \in [0, l]})$ by invoking the algorithm $AIBBE.Encrypt_{IBBE}(M, P_S)$.
- **Decrypt_{or}:** Given the original ciphertext C_S and secret key sk_{id_j} , this algorithm returns the data M' by invoking the algorithm $AIBBE.Decrypt_{IBBE}(C_S, sk_{id_j})$.

4) *Cross-Platoon Authorization:* Assuming the platoon Q wants to merge into P , the PH^P executes the algorithm $Authorize(\mathbb{A}_Q, id_1, sk_{id_1}, P_{S_P})$ to produce an authorization token AT and sends AT to the RSU, in which $\mathbb{A}_Q$ is the access policy that can be matched by the attributes of vehicles in platoon Q . Furthermore, the PH^P sends the verification key vk_{sig}^P to the platoon Q . Using AT , the RSU transforms the ciphertext C_{S_P} disseminated in platoon P into the ciphertext $C_{\mathbb{A}_Q}$ accessible to the vehicles in Q through the algorithm $Transform(AT, C_{S_P})$ and broadcasts $C_{\mathbb{A}_Q}$ to the platoon Q .

- **Authorize:** Given the access policy $\mathbb{A} = (\mathbb{M}, \phi, \Gamma)$, identity id , secret key sk_{id} , and platoon parameters P_S , where

$\mathbb{M}$ is a $n_1 \times n_2$ matrix, ϕ maps each row $\mathbb{M}_i$ of $\mathbb{M}$ into an attribute name, and $\Gamma = (\gamma_{\phi(\mathbb{M}_1)}, \dots, \gamma_{\phi(\mathbb{M}_{n_1})})$ is the attribute value set, this algorithm randomly selects $t, t_1, \dots, t_{n_1} \in \mathbb{Z}_q$ and the vector $\vec{v} \in \mathbb{Z}_q^{n_2}$, and computes $r_i = \mathbb{M}_i \cdot \vec{v}$, for $i \in [1, n_1]$, where $\vec{v} = (s, s_2, \dots, s_{n_2})^T$. Then, the algorithm computes $\chi = \theta^s$, $At_0 = sk_{id} \cdot u^{t \cdot H_4(\chi)}$, $At_1 = P^t$, $At_2 = v^s$, $At_3 = v_1^s$, $At_4 = H_2(\chi || At_1 || At_2 || At_3)$, $At_{5,i} = g_1^{r_i} \cdot o^{t_i} \cdot g^{-t_i}$, $At_{6,i} = v^{t_i \cdot (H_1(2 || \gamma_{\phi(\mathbb{M}_i)) + \beta)}$, for $i \in [1, n_1]$. Finally, it returns the authorization token $AT = ((\mathbb{M}, \phi), At_0, At_1, At_2, At_3, At_4, \{At_{5,i}, At_{6,i}\}_{i \in [1, n_1]})$.

- **Transform:** Given the authorization token AT and original ciphertext C_S , this algorithm calculates $c'_1 = e(At_0, c_1)$, and sets $c'_0 = c_0$, $c'_2 = c_2$, $c'_3 = At_1$, $c'_4 = At_2$, $c'_5 = At_3$, $c'_6 = At_4$, $c'_{7,i} = At_{5,i}$, $c'_{8,i} = At_{6,i}$, $c'_{9,k} = c_{3,k}$, for $i \in [1, n_1]$, $k \in [0, l]$. Finally, the algorithm returns the transformed ciphertext $C_{\mathbb{A}} = ((\mathbb{M}, \phi), c'_0, c'_1, c'_2, c'_3, c'_4, c'_5, c'_6, \{c'_{7,i}, c'_{8,i}\}_{i \in [1, n_1]}, \{c'_{9,k}\}_{k \in [0, l]})$.

5) *Cross-Platoon Access:* Upon receiving $C_{\mathbb{A}_Q}$, the vehicles in Q (i.e., PH and PMs) with the secret key sk_A perform the algorithm $Decrypt_{re}(C_{\mathbb{A}_Q}, sk_A)$ for obtaining the data $M' = m' || sig'$, check the validity and freshness of m' , and execute the corresponding command. If the output of $Decrypt_{re}$ is $\perp$, one of the PMs (e.g., PM_j^Q) executes the algorithm $Claim(C_{\mathbb{A}_Q}, sk_A)$ for generating a proof π and transmits to the TMC, accusing that the RSU didn't generate the transformed ciphertext as required. Then, the TMC runs the algorithm $Verify(C_{\mathbb{A}_Q}, \pi)$ for checking the validity of π and judges whether the RSU performs the algorithm $Transform$ honestly.

- **Decrypt_{re}:** Given the transformed ciphertext $C_{\mathbb{A}}$ and secret key sk_A , this algorithm computes the minimum subset I_{min} satisfying $(\mathbb{M}, \phi)$ from $\{1, \dots, n_1\}$ and finds the coefficients $\{m_i \in \mathbb{Z}_q\}_{i \in I_{min}}$ such that $\sum_{i \in I_{min}} m_i \mathbb{M}_i = (1, 0, \dots, 0)$. Then, it computes χ' by (4) and checks whether $c'_6 = H_2(\chi' || c'_3 || c'_4 || c'_5)$ holds. If it does not hold, the algorithm outputs false, meaning that A fails to match $\mathbb{A}$. Otherwise, it calculates M' and s' by (5), and tests whether $c'_2 = u^{s'}$ or not. If yes, the algorithm outputs M' and $\perp$ otherwise.

$$\chi' = e(sk_1, c'_4^{sk_0} \cdot c'_5) \cdot e\left(\prod_{i \in I_{min}} c'_{7,i}^{m_i}, sk_2\right) \cdot \prod_{i \in I_{min}} e(sk_{3,i}, c'_{8,i})^{m_i}. \quad (4)$$

$$w' = c'_1 / e(c'_2, c'_3)^{H_4(\chi')}, \quad \sigma' = \sum_{k=0}^l c'_{9,k} \cdot H_3(w')^k, \\ M' = c'_0 \oplus H(\sigma'), \quad s' = H_2(M' || \sigma'). \quad (5)$$

- **Claim:** Given the transformed ciphertext $C_{\mathbb{A}}$ and secret key sk_A , this algorithm returns the proof π by running the

¹ Note that in this case of access failure, we require PH^Q to immediately take control of the entire platoon and exit the platoon merge state [60], [61].

algorithm $\text{ZKPoK.Proof}(sk_A, C_A)$, which proves that χ' is correct.

- *Verify*: Given the transformed ciphertext C_A and proof π , this algorithm checks whether $\sum_{i \in I_{min}} m_i M_i = (1, 0, \dots, 0)$ and $c'_6 = H_2(\chi' || c'_3 || c'_4 || c'_5)$ hold. If not, it outputs 0 and aborts. Otherwise, the algorithm computes M' and s' by (5). Then, if $c'_2 = u^{H_2(M' || s')}$ doesn't hold and $\text{true} = \text{ZKPoK.Verify}(\pi, C_A)$, it outputs 1 and 0 otherwise.

The correctness analysis consists of: 1) the decryption of an original ciphertext C_S and transformed ciphertext C_A should success for the identity secret key sk_{id_j} and attribute secret key sk_A if $id_j \in S$ and A matches $\mathbb{A}$; and 2) the verification of proof π should pass if π is honestly produced for an incorrect transformed ciphertext. The detailed correctness analysis can be found in Appendix C, available online.

C. Security Proof

Theorem 1: Our PVCA is IND-CPA-IBBE secure under (P, Q) -GDDHE assumption.

Theorem 2: Our PVCA is IND-CPA-ABE secure under q -PDBDHE assumption.

Theorem 3: Our PVCA is verifiable under CFH assumption.

Theorem 4: Our PVCA is fair if the ZKPoK protocol satisfies soundness property and CFH assumption holds.

Theorem 5: Our PVCA is LR-CA secure under q -SDH assumption.

Proof: The proofs of Theorems 1-5 are given in Appendix D-H, respectively, available online.

D. Security Analysis

A detailed analysis of the security requirements fulfilled by our PVCA is provided in this subsection.

1) *Platoon data confidentiality*: The platoon data is encrypted as the original and transformed ciphertext and then transmitted in the public communication channels. Based on Theorems 1 and 2, any unauthorized vehicles cannot acquire any information from the ciphertext. Consequently, our PVCA achieves confidentiality of platoon data.

2) *Identity and attribute privacy protection*: The identity set of PMs in platoon $\mathbf{P}$ does not need to be transmitted since it is not required for the decryption of the original ciphertext. Meanwhile, the dummy identity rid is embedded in the platoon parameters such that the adversary could not infer the identity set by brute force attacks using c_1 and c_2 . Furthermore, the access policy $\mathbb{A}_Q$ in the authorization token and transformed ciphertext does not contain the attribute value set Γ . Therefore, the identity and attribute privacy of PMs is preserved.

3) *Verifiability and fairness*: In the original ciphertext, the data is incorporated into the random number s , enabling vehicles in platoon $\mathbf{Q}$ to check the correctness of the decryption result with u^s . Meanwhile, accusations against the RSU require the vehicle to produce proof of a valid statement employing its secret key as the witness. Furthermore, according to Theorems 3 and 4, the adversary can neither produce an incorrect transformed ciphertext that could pass verification nor generate a valid proof

for a correct transformed ciphertext. Hence, verifiability and fairness are realized.

4) *Resistance to collusion attack*: On the one hand, in authorization token, the identity secret key sk_{id_1} of PH^P is blinded by the randomness $u^{t \cdot H_4(\chi)}$, i.e., $sk_{id_1} \cdot u^{t \cdot H_4(\chi)}$. Although the collusion of the PMs in platoon $\mathbf{Q}$ and RSU could remove the ABE encapsulated key part $H_4(\chi)$, they still fail to extract sk_{id_1} in the absence of u^t . In addition, Theorem 5 shows that even knowing arbitrary attribute secret keys and authorization tokens, the adversary is unable to deliver a valid sk_{id}^* . On the other hand, Theorems 1 and 2 demonstrate that RSU and PMs cannot access the unauthorized data even if they collude, which is captured by the attribute secret key queries and authorization token queries. Thus, our PVCA provides resistance to a collusion attack.

5) *Resistance to other attacks*: Our PVCA also provides resistance to the following attacks.

- *Replay attack*: The timestamp is included in the protocol data unit field of command data, as shown in Fig. 5. Thus, the PMs in $\mathbf{P}$ and vehicles in $\mathbf{Q}$ can accurately and timely detect replay attacks by checking the freshness of restored data with a predefined threshold and current timestamp.
- *Modification attack*: Assume that the adversary tampers with a data M to M^* , it could to produce the original/transformed ciphertext of M^* . However, after decrypting this ciphertext, the restored data M^* will be discarded since the adversary could not generate a valid signature while lacking sk_{sig}^P .
- *Impersonation attack*: When the adversary impersonates the PH^P , it needs to sign the disseminated command data under vk_{sig}^P . However, it is improbable for the adversary to forge a valid signature without sk_{sig}^P , according to the unforgeability of the underlying DS scheme. Furthermore, based on the mentioned discussion, the adversary cannot impersonate the PMs in $\mathbf{P}$ and vehicles in $\mathbf{Q}$ since it fails to decrypt the received ciphertexts.

E. Further Discussions

This subsection discusses some improvements to our PVCA regarding functionality and security for real-world deployment.

1) *Vehicle Join*: We first consider how to handle the situation of vehicle joining. It is obvious that a new vehicle of the ABE platoon can easily be given access to the transformed ciphertext by issuing a new authorization token. However, adding a new identity id' to the previous IBBE platoon parameters $P_{S_P} = (P, \{P_k\}_{k \in [1, l_P]})$ is not trivial without knowledge of β , where $S_P = \{id_1, \dots, id_{l_P}\}$ is the vehicle identity set. Observing that the exponents of P and P_k could be viewed as the evaluation of β on the polynomial $f_{S_P}(x) = \prod_{j=0}^{l_P} (x + y_j)$, where $y_0 = H_1(0 || rid)$ and $\{y_j = H_1(1 || id_j)\}_{j \in [1, l_P]}$. Therefore, our approach is pre-producing the parameter update material UP such that the PH^P could update the exponents of P and P_k from $f_{S_P}(\beta)$ to $f_{S_P}(\beta) \cdot (\beta + y')$, where $y' = H_1(1 || id')$.

For the IBBE platoon $\mathbf{P}$ with vehicle identity set $S_P = \{id_1, \dots, id_{l_P}\}$, the PH^P will additionally produce $UP = (\{UP_i, \{UP_{k,i}\}_{k \in [1, l_P]}\}_{i \in [1, n - l_P - 1]})$ by (6) when generating

$$P_{S_P} = (P, \{P_k\}_{k \in [1, l_P]}).$$

$$UP_i = P^{\beta^i} = h^{\beta^i \cdot \prod_{j=1}^{l_P} (\beta + H_1(1||id_j)) (\beta + H_1(0||rid))},$$

$$UP_{k,i} = P_k^{\beta^i} = e \left(g, h^{\beta^i \cdot \prod_{j=1, j \neq k}^{l_P} (\beta + H_1(1||id_j)) (\beta + H_1(0||rid))} \right). \quad (6)$$

When l vehicles with the identities $\{id'_1, \dots, id'_l\}$ are allowed to join P ($l < n - l_P$), the vehicle identity set S_P is updated to $S'_P = \{id_1, \dots, id_{l'_P}\}$, in which $l'_P = l_P + l$ and $id_{l_P+j} = id'_j$ for $j \in [1, l]$. In order to permit new vehicles to access the disseminated command data, the PH^P utilizes UP to update the platoon parameters as $P'_{S'_P} = (P', \{P'_k\}_{k \in [1, l'_P]})$ by (7).

$$P' = P \prod_{j=l_P+1}^{l'_P} (\beta + H_1(1||id_j)),$$

$$P'_k = \begin{cases} P_k \prod_{j=l_P+1}^{l'_P} (\beta + H_1(1||id_j)), & k \in [1, l_P] \\ e \left(g, P \prod_{j=l_P+1, j \neq k}^{l'_P} (\beta + H_1(1||id_j)) \right), & k \in [l_P + 1, l'_P] \end{cases} \quad (7)$$

It also renew the parameter update material as $UP' = (\{UP'_i\}_{i \in [1, n-l'_P-1]}, \{UP'_{k,i}\}_{k \in [1, l'_P], i \in [1, n-l'_P-1]})$ by (8).

$$UP'_i = P'^{\beta^i} = h^{\beta^i \cdot \prod_{j=1}^{l'_P} (\beta + H_1(1||id_j)) (\beta + H_1(0||rid))},$$

$$UP'_{k,i} = P'_k^{\beta^i} = e \left(g, h^{\beta^i \cdot \prod_{j=1, j \neq k}^{l'_P} (\beta + H_1(1||id_j)) (\beta + H_1(0||rid))} \right). \quad (8)$$

In this way, the subsequent original ciphertext $C_{S'_P}$ generated using $P'_{S'_P}$ can be decrypted by the new vehicles.

When there are new vehicles permitted to join the ABE platoon Q , the PH^Q updates the access policy $\mathbb{A}_Q$ to $\mathbb{A}'_Q$. Then, the PH^P generates the new authorization token AT' corresponding to $\mathbb{A}'_Q$ and transmits it to the RSU, which will transform the IBBE ciphertext into an ABE ciphertext under $\mathbb{A}'_Q$ using AT' . Consequently, these new vehicles can recover the command data.

2) *Vehicle Leave*: We also discuss the vehicle leaving situation. For the IBBE platoon P , if the PM^P_{leave} with identity id_{leave} intends to leave the platoon, the PH^P only needs to remove the parameters P_{leave} and $\{UP_{leave,i}\}_{i \in [1, n-l_P-1]}$ from P_{S_P} and UP , respectively. Since w_{leave} is missing when forming the polynomial $f(x)$ in the algorithm *Encrypt*, thus decrypting the ciphertext using $sk_{id_{leave}}$ will fail. Therefore, we do not remove the id_{leave} -related information in $(P, \{P_k\}_{k \in [1, l_P], k \neq leave})$. Also, the legacy of id_{leave} in P will guarantee the anonymity of the ciphertext together with rid .

In platoon communication, the broadcast original ciphertext will be received and decrypted by the PMs in platoon P promptly, meaning that there is typically no consideration of revoking a leaving PM from an already-broadcast original ciphertext. However, when necessary, our PVCA can also be amended to support the revocation of leaving PM. Specifically, we transform the generation method of σ from a random selection to an output of a pseudorandom function $F_k(x)$

and allow the PH^P to hold the uniform key k , in which x could be any uniform, random and unduplicated (data-related) information, e.g., the timestamp. Thus, PH^P could quickly infer the secret information (σ, s) for each ciphertext without decryption. Then, the PH^P update $\{c_{3,k}\}_{k \in [0, l]}$ as $\{c'_{3,k} = f'_k\}_{k \in [0, l-1]}$, in which $f'(x) = \frac{f(x) - \sigma}{x - H_3(w_{leave})} + \sigma = \sum_{k=0}^{l-1} f'_k \cdot x^k$, and $w_{leave} = P^s_{leave}$. In this way, the access capability of PM^P_{leave} in the already-broadcast original ciphertexts has been revoked.

For the ABE platoon Q , when the PM^Q_{leave} is going to leave the platoon, the new authorization token is generated for the updated access policy $\mathbb{A}'_Q$, similar to the vehicle join case.

In addition to the above active leaving scenario, we also consider the passive leaving in Q . It is mainly because the secret key in ABE is not associated with individuals, and multiple vehicles may hold the secret key for the same attribute set. Therefore, there exist malicious vehicles that illegally sell their secret keys for economic benefit, while tracking their true identity from the compromised secret keys is difficult. Inspired by the traceable ABE [62], our PVCA can be evolved to support traceability with minor modifications. Specifically, in the algorithm *Setup*(λ), the PKG selects the secure symmetric encryption scheme (SEnc, SDec) with the plaintext spaces $\{0, 1\}^*$ and the key and ciphertext space $\mathbb{Z}_q$, generates the key $\bar{k} \in \mathbb{Z}_q$, and adds $\bar{k}$ into MSK . The algorithm *Extract_{att}* additionally receives the vehicle identity id as input, and the randomness τ (i.e., sk_0 of sk_A) is computed as $\tau = \text{SEnc}(\bar{k}, id||r')$, in which r' is randomly selected from $\mathbb{Z}_q$. In this way, we can append the algorithms (KeySanityCheck and Trace) to our PVCA. The key sanity check algorithm KeySanityCheck is used to check the validity of a compromised secret key, and the trace algorithm identifies the identity of a well-formed secret key.

- *KeySanityCheck*: Given the secret key sk_A , if (9)-(11) hold, this algorithm outputs 1 indicating that sk_A passes the key sanity check and 0 otherwise.

$$sk_0 \in \mathbb{Z}_q \wedge sk_1, sk_2, \{sk_{3,i}\}_{i \in I_A} \in \mathbb{G}, \quad (9)$$

$$e(sk_1, v^{sk_0} \cdot v_1) \cdot e(g_1, sk_2) = \theta, \quad (10)$$

$$\forall i \in I_A, e(sk_{3,i}, v^{H_1(2||a_i)} \cdot v_2) = e(g/o, sk_2) \neq 1_{\mathbb{G}_T}. \quad (11)$$

- *Trace*: Given an well-formed secret key sk_A (i.e., $\text{KeySanityCheck}(sk_A) = 1$) and master secret key MSK , this algorithm computes $id||r' = \text{SDec}(\bar{k}, sk_0)$ and outputs id .

Utilizing the above algorithms, the PKG could track the true identity from a compromised secret key and obtain the corresponding attribute set A . Thus, the access policy $\mathbb{A}_Q$ could be updated to $\mathbb{A}'_Q$, which can not be matched by A , as in the active leaving scenario.

3) *Defending against stronger adversaries*: As proven in Section V-C, our PVCA is secure against chosen plaintext attack (CPA). However, since the active adversaries in the real world may intercept and tamper with the ciphertext transmitted in the public channel and then observe the response of the decryptor, there is a need to consider the stronger chosen ciphertext attack

(CCA), in which the adversaries have the ability to make the decryption queries on the (even replay or malformed) ciphertexts. Under random oracle model, a common method for building CCA-secure encryption schemes is applying Fujisaki-Okamoto transformation [31] to CPA-secure schemes. It is worth noting that our PVCA has employed [31] to support the verifiability of cross-system ciphertext transformation. Therefore, it seems that our PVCA is already able to resist CCA. Nevertheless, as discussed in [32], the support for the ciphertext transformation feature makes it difficult for our PVCA to achieve CCA security of original ciphertexts based on this approach. To tackle this issue, we follow the method of [32] (which introduces the ideas of CHK transform [63] to transform CPA-secure IB-PRE into CCA-secure IB-PRE) by appending a publicly-checkable tag produced through BLS short signature [64] to the original ciphertext, enabling the RSU to verify the validity of ciphertexts and reject malformed ciphertexts. Specifically, the algorithm Encrypt additionally produces $c_4 = H_6(c_0, c_1, c_2, \{c_{3,k}\}_{k \in [0,l]})^s$, where $H_6 : \{0, 1\}^* \rightarrow \mathbb{G}$. In algorithm Transform, the validity of ciphertext is checked through (12). The algorithm continues if it holds and abort otherwise.

$$e(H_6(c_0, c_1, c_2, \{c_{3,k}\}_{k \in [0,l]}), c_2) = e(c_4, u). \quad (12)$$

Through the above approach, our PVCA can be proven to achieve CCA security of original ciphertexts, similar to [32]. Also, the CCA security of transformed ciphertexts has been achieved by adopting Fujisaki-Okamoto transformation. Thus, our PVCA could achieve CCA security at a slight additional overhead, i.e., (a map-to-point hash operation and an exponentiation operation in $\mathbb{G}$ for algorithm Encrypt, 2 pairing operations and a map-to-point hash operation for algorithm Transform, and an element in $|\mathbb{G}|$ for original ciphertext).

4) *Reducing trust in PKG*: Our PVCA assumes that the PKG responsible for initializing the system is completely trusted, i.e., honest-and-isolated PKG. However, the trust anchors in VANETs may be decentralized (i.e., multiple PKG scenarios) and even compromised (i.e., rogue PKG scenarios). Currently, some generic mechanisms have been designed to reduce trust in PKG for IBE and ABE, e.g., distributed PKGs settings [65] which capture an asynchronous communication model among the user and multiple PKGs, and anonymous key issuing protocol [66], [67] which prevents the user's identity from being leaked to a rogue PKG. Consequently, the trust in PKG of our PVCA can be naturally reduced by combining with these mechanisms.

VI. PERFORMANCE EVALUATION

The comprehensive performance evaluation of our PVCA is conducted in this section to demonstrate its practicality.

A. Theoretical Analysis

The comparison results between the well-performing IBE scheme (BF [8]) in real-time communication, the cross-system ciphertext transformation schemes (WLY+ [19], ZYL+ [20], ZFC+ [21]) between IBE and ABE, the VF-AB-PRE scheme

(GSB+ [28]), and our PVCA regarding computation and storage/communication complexity are given in Tables II and III, respectively.

From Table II, the computation complexity of algorithm Encrypt in BF [8], WLY+ [19], ZYL+ [20], and ZFC+ [21] is independent of n_1 or l , but GSB+ [28] and our scheme is relevant due to the fact that the original ciphertext corresponds to multiple receivers. In all schemes, the computation complexity of algorithm Authorize is proportional to n_1 , while WLY+ [19] and GSB+ [28] additionally correlate with n_2 and n , respectively. Note that the computation complexity of algorithms Transform and Decrypt_{or} in all schemes other than GSB+ [28] remains stable. Regarding the algorithm Decrypt_{tr}, GSB+ [28] and our PVCA have lower computation complexity compared with ZYL+ [20] and ZFC+ [21].

It is clear from Table III that the storage complexity of identity secret key in PVCA is the same as the classical IBE scheme [8], and is superior to WLY+ [19], ZYL+ [20], and ZFC+ [21]. The storage complexity of attribute secret key of our PVCA is comparable to ZFC+ [21] and GSB+ [28], and outperforms WLY+ [19] and ZYL+ [20]. Furthermore, the storage complexity of authorization token in PVCA is the smallest compared to other schemes. Since the original ciphertext in our PVCA corresponds to multiple receivers, the communication complexity of original and transformed ciphertext naturally rises as l increases.

B. Evaluation Settings

We implement a prototype of our PVCA to evaluate the efficiency, in which a desktop computer and Raspberry Pi 4 Model B are utilized to emulate the PKG/TMC/RSUs and resource-constrained vehicles, respectively. Meanwhile, the average packet latency and throughput of our PVCA for platoon communication in VANETs are simulated to analyze performance from a network point of view. Also, we evaluate the RSU serving rate to demonstrate the scalability of PVCA. We list the specific environmental details in Table IV.

For prototype evaluation, the Type A pairing on curve $E(\mathbb{F}_p) : y^2 = x^3 + x$ is selected under 128-bit security level, where $|\mathbb{G}|$, $|\mathbb{G}_T|$, and $|\mathbb{Z}_q|$ are 3072, 3072, and 256 bits, respectively. Furthermore, we instantiate the DS protocol utilizing ECDSA signature, which is implemented under the secp256k1 elliptic curve. Thus, the size of data $|M|$ is 149 bytes. Meanwhile, input and output files for all algorithms are in XML format. Based on AES in the counter mode, random tapes are pseudorandomly produced, in which the keys are generated through the secure randomness generator of PyCrypto. In this way, obtaining a randomness of 256 bits takes about 0.14 ms (including deserialization to group), resulting in a PRNG speed of 223.21 KB/s.

For simulation evaluation, we deploy a virtual bidirectional eight-lane highway scenario. The specific parameter settings are demonstrated in Table V.

C. Prototype Evaluation

Fig. 6 depicts the efficiency comparison results between our PVCA and the schemes [8], [19], [20], [21], [28].

TABLE II
COMPARISON OF COMPUTATION COMPLEXITY

Schemes	Encrypt	Authorize	Transform	Decrypt _{or}	Decrypt _{tr}
BF [8]	$P + H + E_T + E$	—	—	P	—
WLY+ [19]	$6H + 2E_T + 12E$	$(6n_1 + 6n_2 + 3)H + 2E_T + (6n_1n_2 + 6n_1 + 9)E$	$6P$	$6P$	$9P + 3H$
ZYL+ [20]	$E_T + 4E$	$E_T + (5n_1 + 3)E$	$2P$	$2P$	$(2n + 3)P + H + n(E_T + E)$
ZFC+ [21]	$E_T + 4E$	$E_T + (3n_1 + 2)E$	$2P$	—	$(n + 3)P + HnE_T + (3n + 1)E$
GSB+ [28]	$E_T + (3n_1 + 4)E$	$2E_T + (3n_1 + n + 5)E$	$(2n_1 + 2)P + n_1E_T$	$nE_T + (n + 2)(P + E)$	$(n + 1)E_T + (n + 2)(P + E)$
Ours	$lE_T + 2E$	$E_T + (4n_1 + 4)E$	P	$P + E$	$(n + 3)P + (n + 1)E_T + (n + 2)E$

P : Time of bilinear pairing operation; H : Time of map-to-point hash operation; E_T, E : Time of exponentiation operation in $\mathbb{G}_T, \mathbb{G}$; l : Size of identity set; $n_1 \times n_2$: Size of access policy; n : Size of attribute set; “—”: Not applicable.

TABLE III
COMPARISON OF STORAGE/COMMUNICATION COMPLEXITY

Schemes	Identity Secret Key	Attribute Secret Key	Authorization Token	Original Ciphertext	Transformed Ciphertext
BF [8]	$ \mathbb{G} $	—	—	$ \mathbb{G} + M $	—
WLY+ [19]	$6 \mathbb{G} $	$(3n + 6) \mathbb{G} $	$(3n_1 + 12) \mathbb{G} $	$ \mathbb{G}_T + 9 \mathbb{G} $	$ \mathbb{G}_T + (3n_1 + 9) \mathbb{G} $
ZYL+ [20]	$2 \mathbb{G} $	$(2n + 2) \mathbb{G} $	$(3n_1 + 4) \mathbb{G} $	$ \mathbb{G}_T + 3 \mathbb{G} $	$ \mathbb{G}_T + (3n_1 + 3) \mathbb{G} $
ZFC+ [21]	$2 \mathbb{G} $	$(n + 2) \mathbb{G} $	$(2n_1 + 4) \mathbb{G} + (2n_1 + 1) \mathbb{Z}_q $	$ \mathbb{G}_T + 3 \mathbb{G} $	$ \mathbb{G}_T + (2n_1 + 3) \mathbb{G} + (2n_1 + 1) \mathbb{Z}_q $
GSB+ [28]	—	$(n + 2) \mathbb{G} $	$2 \mathbb{G}_T + (n + 2n_1 + 4) \mathbb{G} $	$(2n_1 + 3) \mathbb{G} + 2 M $	$3 \mathbb{G}_T + (2n_1 + 2) \mathbb{G} + 2 M $
Ours	$ \mathbb{G} $	$(n + 2) \mathbb{G} + \mathbb{Z}_q $	$(2n_1 + 4) \mathbb{G} + \mathbb{Z}_q $	$2 \mathbb{G} + l \mathbb{Z}_q + M $	$ \mathbb{G}_T + (2n_1 + 4) \mathbb{G} + (l + 1) \mathbb{Z}_q + M $

$|\mathbb{G}|, |\mathbb{G}_T|, |\mathbb{Z}_q|$: Size of element in $\mathbb{G}, \mathbb{G}_T$, and $\mathbb{Z}_q$; $|M|$: Size of data.

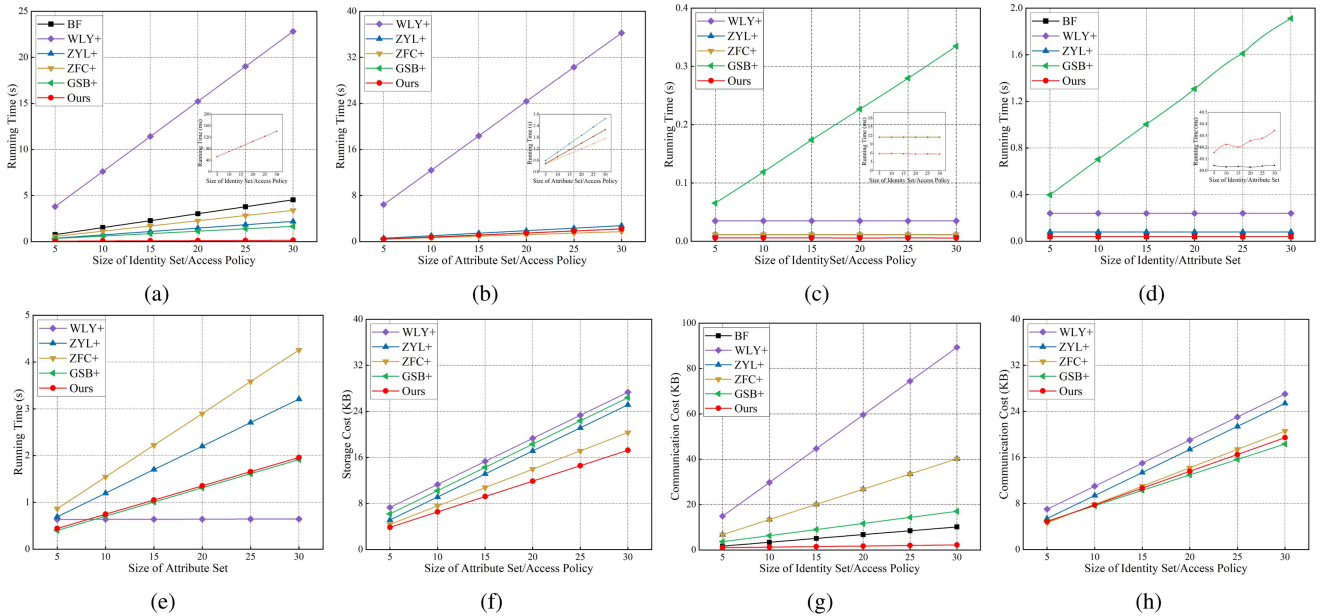


Fig. 6. Comparison of efficiency between our PVCA and the schemes [8], [19], [20], [21], [28]. (a) Computation cost of Encrypt. (b) Computation cost of Authorize. (c) Computation cost of Transform. (d) Computation cost of Decrypt_{or}. (e) Computation cost of Decrypt_{tr}. (f) Storage cost of authorization token. (g) Communication cost of original ciphertext. (h) Communication cost of transformed ciphertext.

Fig. 6(a) illustrates the running time of algorithm Encrypt in BF [8], WLY+ [19], ZYL+ [20], ZFC+ [21], GSB+ [28], and our scheme. It can be seen that all schemes are both proportional to l or n_1 , while our PVCA rises more slowly than other schemes. Specifically, when $l = n_1 = 30$, PVCA merely spends 0.14s to encrypt the command data, but BF [8], WLY+ [19], ZYL+ [20], ZFC+ [21], and GSB+ [28] need 4.54, 22.82, 2.19, 3.39, and 1.66s, respectively. Thus, compared to the original ciphertext generation time of other schemes, PVCA achieves a saving of at least 91.57%. This is mainly because the algorithm Encrypt

in BF [8], WLY+ [19], ZYL+ [20], and ZFC+ [21] have to be performed l times to generate the original ciphertexts for the identity set, while our scheme only needs to be executed once. Our PVCA also reduces the computation cost by pre-generating the platoon parameters.

From Fig. 6(b), the running time of algorithm Authorize is linear with n_1, n_2 , or n . Clearly, our scheme achieves a comparable time consumption to GSB+ [28] and better efficiency than WLY+ [19] and ZYL+ [20]. When $n_1 = n_2 = n = 30$, PVCA needs 2.19s to generate the authorization token, while 36.25 and

TABLE IV
ENVIRONMENTAL DETAILS

	Environment	Details
Desktop computer	CPU	Intel Core i5-10400 (@ 2.90GHz)
	RAM	8 GB
	External storage	100 GB
	Operating system	Ubuntu 18.04.6 LTS
Raspberry Pi	CPU	ARM Cortex-A72 (up to 1.80GHz)
	RAM	4 GB
	External storage	32 GB
	Operating system	Debian GNU/Linux 12
Other software	Programming language	Python 3.6 & OTcl-1.14
	Program library	Charm-Crypto 0.5 & PyCrypto 2.6.1 & NumPy 1.19.5
	Network simulator	NS-2.35
	Data analysis tool	Gawk-1:4.1.4

TABLE V
PARAMETER SETTINGS FOR SIMULATION EVALUATION

Parameter	Setting	Parameter	Setting
Road length	10 km	Vehicle length	5 m
Network protocol	DSRC	Default speed	100 km/h
Network bandwidth	24 Mbps	Default distance	14 m
Simulation time	100 s	Command dissemination period	200 ms
Communication range of RSU	600 m	Communication range of vehicle	300 m
Number of RSU	23	Number of vehicle	5 ~ 20

2.77s are required in WLY+ [19] and ZYL+ [20], respectively. That is, PVCA could reduce the authorization token generation time by 93.96% and 20.94% than WLY+ [19] and ZYL+ [20], respectively. Although ZFC+ [21] has slightly higher efficiency compared to our PVCA, it ignores the attribute privacy and verifiability.

The running time of algorithms Transform is exhibited in Fig. 6(c). It is evident that WLY+ [19], ZYL+ [20], ZFC+ [21], and our PVCA obtain the constant cost and are superior to GSB+ [28], of which the running time naturally rises as n_1 increases. Meanwhile, our scheme outperforms the schemes [19], [20], [21]. Concretely, when $l = n_1 = 30$, the conversion of the IBBE ciphertext to an ABE ciphertext in PVCA requires only 5.70 ms, compared to 34.98, 11.37, 11.73, and 334.86 ms in WLY+ [19], ZYL+ [20], ZFC+ [21], and GSB+ [28], respectively. Hence, PVCA represents a minimum time saving of 49.87% over the other schemes.

Fig. 6(d) provides the running time of algorithms Decrypt_{or}. It is critical to note that the original ciphertext decrypted in GSB+ [28] and our PVCA is a multi-receiver ciphertext, in contrast to the single-receiver ciphertext used in [8], [19], [20], [21]. Nonetheless, our PVCA exhibits a comparable performance to BF [8] and clear superiority over other schemes supporting ciphertext transformation. Specifically, PVCA just requires 0.04s to decrypt the original ciphertext with $l = n = 30$, while WLY+ [19], ZYL+ [20], ZFC+ [21], and GSB+ [28] need 0.24, 0.08, 3.39, and 1.91s, respectively. Therefore, PVCA achieves a reduction in original ciphertext decryption time of at least 50% compared to the schemes [19], [20], [21], [28].

Fig. 6(e) describes the running time of algorithm Decrypt_{tr} in WLY+ [19], ZYL+ [20], ZFC+ [21], GSB+ [28], and our PVCA.

It is notable that PVCA has less running time than ZYL+ [20] and ZFC+ [21]. To be specific, when $n = 30$, PVCA requires 1.96 s to decrypt the transformed ciphertext, which is prominently faster than the 3.21 s in ZYL+ [20] and the 4.26 s in ZFC+ [21], implying a performance gain of approximately 38.94% and 53.99%, respectively. Since WLY+ [19] adopts the fast ABE scheme (i.e., FAME) to generate the authorization token, its running time remains stable. Also, the running time of GSB+ [28] is slightly outperformed by our scheme. Nevertheless, WLY+ [19] and GSB+ [28] either fail to achieve verifiability and fairness or cannot support anonymity and a large universe. Meanwhile, we can also borrow the ideas of edge computing and outsourced decryption to reduce the transformed ciphertext decryption cost, i.e., offloading the decryption task to the RSU, similar to [33], [44].

As demonstrated in Fig. 6(f) and (g), the storage cost of authorization token and communication cost of original ciphertext in our PVCA are the lowest. When $n_1 = n = 30$, the size of authorization token in PVCA is 17.24 KB, but that of WLY+ [19], ZYL+ [20], ZFC+ [21], and GSB+ [28] are 27.33, 25.15, 20.33, and 26.39 KB, respectively. Hence, PVCA takes at least 15.20% less storage for authorization tokens compared to other schemes. Since the original ciphertext in BF [8], WLY+ [19], ZYL+ [20], and ZFC+ [21] is an IBE ciphertext, thus their communication costs far exceed that of GSB+ [28] (ABE ciphertext) and our PVCA (IBBE ciphertext) for multi-receiver scenarios. Meanwhile, the adoption of polynomial technique makes the size of original ciphertext in our scheme dramatically shorter. Specifically, when $l = n_1 = 30$, the original ciphertext in PVCA has 2.23 KB, while that in BF [8], WLY+ [19], ZYL+ [20], ZFC+ [21], and GSB+ [28] has 10.17, 89.33, 40.20, 40.20, and 17.04 KB, respectively. Namely, for platoon data dissemination, PVCA delivers no less than a 78.07% lower communication cost.

Fig. 6(h) shows the communication cost of transformed ciphertext. It is noteworthy that transformed ciphertext in PVCA has a lower size compared with WLY+ [19] and ZYL+ [20]. The communication cost in ZFC+ [21], GSB+ [28], and our scheme is similar. However, they do not provide support for a large universe and attribute privacy protection.

Table VI also reports more specific running time of all algorithms and sizes of output files in our PVCA, where “Input from disk”, “Algorithm exec”, and “Output to disk” represent the running time of reading the input parameters from file, executing the algorithm, and writing the algorithm output to disk, respectively. It can be seen that the time spent reading the input or writing the output is much lower than that spent executing the algorithms. Also, due to limited computational and storage resources, the running time of algorithms executed in Raspberry Pi (i.e., Establish, Encrypt, Decrypt_{or}, Authorize, Decrypt_{re}, and Claim) is significantly higher than that in the desktop computer.

D. Simulation Evaluation

Fig. 7 shows the simulation results of our PVCA. We consider the two communication modes: platoon data dissemination and cross-platoon access. Note that the NS-2 simulator primarily

TABLE VI
RUNNING TIME OF ALGORITHMS AND SIZES OF OUTPUT FILES

	$l/n_1/n$	Setup	Extract _{id}	Extract _{att}	Establish	Encrypt	Decrypt _{or}	Authorize	Transform	Decrypt _{tr}	Claim	Verify
Input from disk (ms)	5	—	0.02	0.02	0.05	0.10	0.08	0.14	0.02	0.10	0.16	0.04
	10	—	0.03	0.02	0.06	0.11	0.08	0.15	0.05	0.11	0.17	0.04
	15	—	0.02	0.02	0.06	0.11	0.08	0.15	0.03	0.14	0.19	0.05
	20	—	0.02	0.03	0.07	0.13	0.08	0.17	0.03	0.15	0.22	0.05
	25	—	0.03	0.03	0.07	0.14	0.09	0.18	0.03	0.16	0.23	0.06
	30	—	0.03	0.03	0.07	0.15	0.09	0.19	0.04	0.18	0.25	0.07
Algorithm exec (ms)	5	66.52		32.96	745.15	52.75	40.16	429.53	5.76	441.79	791.94	121.93
	10	89.34		56.41	2347.80	70.56	40.22	782.12	5.81	746.69	1270.97	175.94
	15	115.27	4.72	79.94	4821.33	88.06	40.20	1135.47	5.79	1049.87	1749.46	231.23
	20	138.68		103.38	8180.46	105.86	40.26	1487.95	5.70	1352.93	2226.02	283.69
	25	162.36		126.65	12421.80	123.41	40.28	1840.30	5.72	1656.69	2699.40	340.45
	30	186.18		149.85	17535.74	141.16	40.34	2192.88	5.70	1960.22	3178.54	392.60
Output to disk (ms)	5	0.39		0.39	1.22	0.80	—	1.18	0.27	—	0.85	—
	10	0.35		0.30	1.70	1.06	—	1.75	0.28	—	1.67	—
	15	0.51	0.23	0.32	1.37	1.22	—	1.66	0.29	—	1.73	—
	20	0.35		0.31	1.75	1.28	—	1.49	0.43	—	0.90	—
	25	0.32		0.26	2.10	1.34	—	2.22	0.40	—	1.28	—
	30	0.46		0.26	2.19	1.32	—	2.03	0.33	—	1.68	—
Output file size (bytes)	5	4640		2026	2905	1033	—	3954	4956	—	3138	—
	10	5985		3411	5510	1283	—	6694	7946	—	4523	—
	15	7330	288	4796	8115	1533	—	9434	10936	—	5908	—
	20	8675		6181	10720	1783	—	12174	13926	—	7293	—
	25	10020		7566	13325	2033	—	14914	16916	—	8678	—
	30	11365		8995	15930	2283	—	17658	19910	—	10067	—

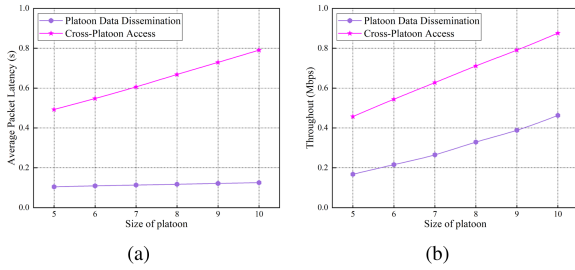


Fig. 7. Simulation of our PVCA. (a) Average packet latency. (b) Throughput.

focuses on evaluating network-level performance, i.e., the exchange of packets, without considering the latency that may arise when processing real sensor data. In reality, the processing, signing, and encrypting of command data will introduce significant computational overhead, potentially impacting overall end-to-end latency and throughput. Consequently, to simulate real-world scenarios as closely as possible, we consider the processing latency in platoon data dissemination includes the signature generation time, data encryption time, original ciphertext decryption time, and signature verification time, while that in cross-platoon access contains the signature generation time, data encryption time, transformation time, transformed ciphertext decryption time, and signature verification time. Also, we obtain the specific values for these times under different platoon sizes through the implemented prototype.

The average packet latency under different platoon sizes is depicted in Fig. 7(a). It can be seen that average packet latency for cross-platoon access grows with increasing the size of platoon while that for platoon data dissemination rises exceedingly slowly. The main reason is that the transmission latency is completely influenced by the packet length when the bandwidth is fixed. Compared to cross-platoon access, the packet length for platoon data dissemination is shorter and less

variable. Meanwhile, cross-platoon access has a longer processing latency than platoon data dissemination since it contains non-constant decryption time and additional transformation time. Furthermore, a higher propagation latency is caused by the fact that the RSU must process the packet for cross-platoon access.

Fig. 7(b) illustrates the throughput under different platoon sizes. Obviously, as the platoon size grows larger, the throughput for platoon data dissemination and cross-platoon access both increase since the number of receivers for the packet expands. Also, cross-platoon access achieves higher throughput compared to platoon data dissemination since the length of a packet containing the transformed ciphertext is much larger than that of a packet containing the original ciphertext. Specifically, when the platoon size is 10, the throughput for platoon data dissemination is 0.46 Mbps, while that for cross-platoon access attains 0.88 Mbps. From the above analysis, our PVCA is feasible for platoon communication in VANETs.

E. RSU Serving Rate Evaluation

We consider the capability of the RSU to handle ciphertext transformation in our PVCA (i.e., scalability of the scheme). As the number of platoons and the proportion of merged platoons increase within the communication range, the RSU may fail to transform the original ciphertexts in a timely manner, leading to data loss. Let N_{pro} and N_{rec} denote the number of original ciphertexts that the RSU can process per unit time and the number of original ciphertexts received by the RSU per unit time, respectively. The RSU serving rate can be expressed by (13).

$$R_{ser} = \begin{cases} 1, & \text{if } N_{pro} > N_{rec} \\ N_{pro}/N_{rec}, & \text{otherwise} \end{cases}. \quad (13)$$

Define t as the processing time (i.e., performing ciphertext transformation) required by the RSU after receiving the original

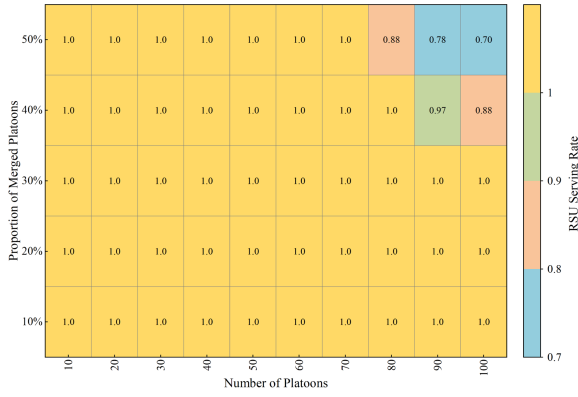


Fig. 8. RSU serving rate.

ciphertext, and p_c as the command dissemination period of platoons. We configure each platoon to consist of five vehicles. According to the communication range of RSU, vehicle length, and the default distance in evaluation settings, it can be obtained that the maximum number of platoons within the service range of an RSU is approximately 100. Thus, we denote n_p by the number of platoons which varies from 10 to 100 and p_m by the proportion of merged platoons in all platoons which varies from 10% to 50%. Therefore, the RSU serving rate can be calculated as (14).

$$R_{ser} = \begin{cases} 1, & \text{if } 1/t > p_m \cdot n_p / p_c \\ p_c / (p_m \cdot n_p \cdot t), & \text{otherwise} \end{cases} \quad (14)$$

Based on the processing time obtained from the implemented prototype, Fig. 8 shows the RSU serving rate under different n and m . It can be observed that with increasing number of platoons and the proportion of merged platoons, the RSU serving rate gradually decreases. The primary reason lies in the fact that an increase in the number of merged platoons leads to the RSU performing more ciphertext transformation operations. However, except in extreme cases where the proportion of merged platoons reaches 50% and traffic becomes extremely congested, the RSU serving rate can be maintained at 1. Consequently, we can conclude that the RSU in PVCA is unlikely to become overloaded due to the massive ciphertext transformation tasks.

VII. CONCLUSION

In this paper, we constructed a privacy-preserving and verifiable cross-system authorization scheme for platoon communications, where direct authorization is enabled by transforming ciphertexts of IBBE format into new ciphertexts of ABE format. With the optimized AIBBE and PH-ABE protocols, the identity and attribute privacy of vehicles can be protected in the original and transformed ciphertext, respectively. Furthermore, to prevent the untrusted proxy from incorrectly transforming ciphertexts and the honest proxy from being maliciously accused, the proposed scheme achieved more practical cross-system ciphertext transformation verification by designing the ZKPoK protocol. The formal security proof demonstrated that our scheme is secure. The comprehensive performance evaluation indicated that, our scheme provides superior efficiency with

better suitability for platoon communications, compared to the related schemes. Although the proposed scheme offers higher security and efficiency, the ciphertext size grows linearly with the size of identity set and access policy. For future work, we plan to construct a more practical cross-system authorization scheme for large-scale platoon communications to achieve constant or sub-linear ciphertext growth.

REFERENCES

- [1] P. Varaiya, "Smart cars on smart roads: Problems of control," *IEEE Trans. Autom. Control*, vol. 38, no. 2, pp. 195–207, Feb. 1993.
- [2] X. Lin, R. Lu, C. Zhang, H. Zhu, P.-h. Ho, and X. Shen, "Security in vehicular ad hoc networks," *IEEE Commun. Mag.*, vol. 46, no. 4, pp. 88–95, Apr. 2008.
- [3] A. A. Alam, A. Gattami, and K. H. Johansson, "An experimental study on the fuel reduction potential of heavy duty vehicle platooning," in *Proc. Int. IEEE Conf. Intell. Transp. Syst.*, 2010, pp. 306–311.
- [4] C. Lai, R. Lu, and D. Zheng, "SPGS: A secure and privacy-preserving group setup framework for platoon-based vehicular cyber-physical systems," *Secur. Commun. Netw.*, vol. 9, no. 16, pp. 3854–3867, 2016.
- [5] D. Xu et al., "Post-quantum authentication against cyber-physical attacks in V2X-based autonomous vehicle platoon," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 6, pp. 5034–5044, Jun. 2024.
- [6] R. Lu, X. Lin, H. Zhu, P.-H. Ho, and X. Shen, "ECPP: Efficient conditional privacy preservation protocol for secure vehicular communications," in *Proc. IEEE Conf. Comput. Commun.*, 2008, pp. 1229–1237.
- [7] H. Liu, Y. Ming, C. Wang, Y. Zhao, S. Zhang, and R. Lu, "Server-assisted data sharing system supporting conjunctive keyword search for vehicular social networks," *IEEE Trans. Serv. Comput.*, vol. 17, no. 6, pp. 4281–4295, Nov./Dec. 2024.
- [8] D. Boneh and M. Franklin, "Identity-based encryption from the weil pairing," in *Proc. Adv. Cryptology-CRYPTO'01*, 2001, pp. 213–229.
- [9] J. Bethencourt, A. Sahai, and B. Waters, "Ciphertext-policy attribute-based encryption," in *Proc. IEEE Symp. Secur. Privacy*, 2007, pp. 321–334.
- [10] C. Delerablée, "Identity-based broadcast encryption with constant size ciphertexts and private keys," in *Proc. Adv. Cryptology-ASIACRYPT'07*, 2007, pp. 200–215.
- [11] T. Matsuo, "Proxy re-encryption systems for identity-based encryption," in *Proc. Int. Conf. Pairing-Based Cryptogr.*, 2007, pp. 247–267.
- [12] P. Jiang, J. Ning, K. Liang, C. Dong, J. Chen, and Z. Cao, "Encryption switching service: Securely switch your encrypted data to another format," *IEEE Trans. Serv. Comput.*, vol. 14, no. 5, pp. 1357–1369, Sep./Oct. 2021.
- [13] H. Deng, Q. Wu, B. Qin, W. Susilo, J. Liu, and W. Shi, "Asymmetric cross-cryptosystem re-encryption applicable to efficient and secure mobile access to outsourced data," in *Proc. ACM Symp. Inf. Comput. Commun. Secur.*, 2015, pp. 393–404.
- [14] J. Sun, H. Xiong, H. Zhang, and L. Peng, "Mobile access and flexible search over encrypted cloud data in heterogeneous systems," *Inf. Sci.*, vol. 507, pp. 1–15, 2020.
- [15] H. Deng et al., "Identity-based encryption transformation for flexible sharing of encrypted data in public cloud," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 3168–3180, 2020.
- [16] J. Sun, G. Xu, T. Zhang, X. Cheng, X. Han, and M. Tang, "Secure data sharing with flexible cross-domain authorization in autonomous vehicle systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 7, pp. 7527–7540, Jul. 2023.
- [17] Y. Zhao, H. Yu, Y. Yang, L. Xu, S. Pan, and Y. Ren, "Flexible and secure cross-domain signcrypted data authorization in multi-platoon vehicular networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 3, pp. 3166–3176, Mar. 2024.
- [18] K. He et al., "A new encrypted data switching protocol: Bridging IBE and ABE without loss of data confidentiality," *IEEE Access*, vol. 7, pp. 50658–50668, 2019.
- [19] L. Wang, Y. Lin, T. Yao, H. Xiong, and K. Liang, "FABRIC: Fast and secure unbounded cross-system encrypted data sharing in cloud computing," *IEEE Trans. Dependable Secure Comput.*, vol. 20, no. 6, pp. 5130–5142, Nov./Dec. 2023.
- [20] Y. Zhao, H. Yu, Y. Liang, H. Jiang, G. Marine, and Y. Ren, "Sanitizable cross-system authorization for secure communication in intelligent connected vehicle," *IEEE Trans. Veh. Technol.*, vol. 72, no. 11, pp. 15002–15013, Nov. 2023.

- [21] Q. Zhang, Y. Fu, J. Cui, D. He, and H. Zhong, "Efficient fine-grained data sharing based on proxy re-encryption in IIoT," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 6, pp. 5797–5809, Nov./Dec. 2024.
- [22] M. Blaze, G. Bleumer, and M. Strauss, "Divertible protocols and atomic proxy cryptography," in *Proc. Adv. Cryptology-EUROCRYPT'98*, 1998, pp. 127–144.
- [23] S. Maiti and S. Misra, "P2B: Privacy preserving identity-based broadcast proxy re-encryption," *IEEE Trans. Veh. Technol.*, vol. 69, no. 5, pp. 5610–5617, May 2020.
- [24] J. Sun, G. Xu, T. Zhang, X. Yang, M. Alazab, and R. H. Deng, "Verifiable, fair and privacy-preserving broadcast authorization for flexible data sharing in clouds," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 683–698, 2023.
- [25] J. Zhang, S. Su, H. Zhong, J. Cui, and D. He, "Identity-based broadcast proxy re-encryption for flexible data sharing in VANETs," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 4830–4842, 2023.
- [26] Y. Zhang, J. Li, X. Chen, and H. Li, "Anonymous attribute-based proxy re-encryption for access control in cloud computing," *Secur. Commun. Netw.*, vol. 9, no. 14, pp. 2397–2411, 2016.
- [27] J. Zhao, K. Zhang, J. Gong, and H. Qian, "Lavida: Large-universe, verifiable, and dynamic fine-grained access control for e-health cloud," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 2732–2745, 2024.
- [28] C. Ge, W. Susilo, J. Baek, Z. Liu, J. Xia, and L. Fang, "A verifiable and fair attribute-based proxy re-encryption scheme for data sharing in clouds," *IEEE Trans. Dependable Secure Comput.*, vol. 19, no. 5, pp. 2907–2919, Sep./Oct. 2022.
- [29] Z. Song, H. Ma, R. Zhang, W. Xu, and J. Li, "Everything under control: Secure data sharing mechanism for cloud-edge computing," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 2234–2249, 2023.
- [30] L. Jiang, M. Alazab, and Z. Qin, "Secure task distribution with verifiable re-encryption in mobile crowdsensing assisted emergency IoT system," *IEEE Internet Things J.*, vol. 11, no. 3, pp. 3896–3908, Feb. 2024.
- [31] E. Fujisaki and T. Okamoto, "Secure integration of asymmetric and symmetric encryption schemes," in *Proc. Adv. Cryptology-CRYPTO'99*, 1999, pp. 537–554.
- [32] M. Green and G. Ateniese, "Identity-based proxy re-encryption," in *Proc. Int. Conf. Appl. Cryptogr. Netw. Secur.*, 2007, pp. 288–306.
- [33] H. Liu, Y. Ming, C. Wang, and Y. Zhao, "Flexible selective data sharing with fine-grained erasure in VANETs," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 9582–9597, 2024.
- [34] F. Zhang, Y. Zhang, Z. Ma, L. Sun, and Y. Shen, "A regulatable blockchain rewriting scheme for identity-aware data modification and user identity update," *ACM Trans. Web.*, vol. 20, 2025, Art. no. 14.
- [35] C. Li, R. Chen, Y. Wang, Q. Xing, and B. Wang, "REEDS: An efficient revocable end-to-end encrypted message distribution system for IoT," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 5, pp. 4526–4542, Sep./Oct. 2024.
- [36] P. Xu, T. Jiao, Q. Wu, W. Wang, and H. Jin, "Conditional identity-based broadcast proxy re-encryption and its application to cloud email," *IEEE Trans. Comput.*, vol. 65, no. 1, pp. 66–79, Jan. 2016.
- [37] C. Ge, Z. Liu, J. Xia, and L. Fang, "Revocable identity-based broadcast proxy re-encryption for data sharing in clouds," *IEEE Trans. Dependable Secure Comput.*, vol. 18, no. 3, pp. 1214–1226, May/Jun. 2021.
- [38] L. Chen, M. Zhang, and J. Li, "Conditional identity-based broadcast proxy re-encryption with anonymity and revocation," *IEEE Trans. Reliab.*, vol. 74, no. 3, pp. 3573–3584, Sep. 2025.
- [39] L. Chen, S. Chen, H. Zhang, and J. Weng, "Fair and verifiable identity-based broadcast proxy re-encryption with designated sender feasible for medical Internet of Things," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 6033–6045, 2025.
- [40] X. Liang, Z. Cao, H. Lin, and J. Shao, "Attribute based proxy re-encryption with delegating capabilities," in *Proc. ACM Symp. Inf., Comput. Commun. Secur.*, 2009, pp. 276–286.
- [41] Y. Zhang, D. Li, X. Zhang, and W. Lv, "An efficient cross-domain fine grain proxy re-encryption scheme for secure transmission in IIOT," *ACM Trans. Sensor Netw.*, vol. 21, no. 3, pp. 1–21, 2025.
- [42] F. Zhao, J. Weng, W. Xie, L. Hou, and M. Li, "Time-based attribute-based proxy re-encryption with decryption key update," *Designs Codes Cryptogr.*, vol. 92, pp. 4099–4129, 2024.
- [43] P. Duan, Z. Ma, H. Gao, T. Tian, and Y. Zhang, "Multi-authority attribute-based encryption scheme with access delegation for cross blockchain data sharing," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 323–337, 2025.
- [44] H. Zhong et al., "Security-enhanced data sharing via efficient sanitization for VANETs," *IEEE Trans. Mobile Comput.*, vol. 24, no. 6, pp. 4925–4938, Jun. 2025.
- [45] F. Jiang, B. Qi, T. Wu, K. Zhu, and L. Zhang, "CPSS: CP-ABE based platoon secure sensing scheme against cyber-attacks," in *Proc. IEEE Intell. Transp. Syst. Conf.*, 2019, pp. 3218–3223.
- [46] C. Lai, G. Li, and D. Zheng, "SPSC: A secure and privacy-preserving autonomous platoon setup and communication scheme," *Trans. Emerg. Telecommun. Technol.*, vol. 32, no. 9, 2021, Art. no. e3982.
- [47] Y. Zhao, Y. Wang, X. Cheng, H. Chen, H. Yu, and Y. Ren, "RFAP: A revocable fine-grained access control mechanism for autonomous vehicle platoon," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 9668–9679, Jul. 2022.
- [48] Y. Zhao, Y. Wang, Y. Liang, H. Yu, and Y. Ren, "Identity-based broadcast signcryption scheme for vehicular platoon communication," *IEEE Trans. Ind. Informat.*, vol. 19, no. 6, pp. 7814–7824, Jun. 2023.
- [49] Y. Bao et al., "Lightweight and bilateral controllable data sharing for secure autonomous vehicles platooning service," *IEEE Trans. Veh. Technol.*, vol. 72, no. 11, pp. 13969–13984, Nov. 2023.
- [50] Y. Zhao et al., "Secure source identification scheme for revocable instruction sharing in vehicle platoon," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 5, pp. 3671–3684, May 2024.
- [51] G. Wang, L. Wang, Z. Liu, K. Zhang, M. He, and W. Li, "An innovative blockchain-assisted multi-message and multi-receiver heterogeneous signcryption for cross-domain vehicle platoon communication," *IEEE Trans. Veh. Technol.*, vol. 75, no. 4, pp. 5892–5907, Apr. 2026.
- [52] C. Zhang et al., "TPPR: A trust-based and privacy-preserving platoon recommendation scheme in VANET," *IEEE Trans. Serv. Comput.*, vol. 15, no. 2, pp. 806–818, Mar./Apr. 2022.
- [53] S. Li et al., "TD-SAS: A trust-aware and decentralized speed advisory system for energy-efficient autonomous vehicle platoons," *IEEE Trans. Intell. Veh.*, vol. 9, no. 9, pp. 5554–5570, Sep. 2024.
- [54] R. Li et al., "RPPM: A reputation-based and privacy-preserving platoon management scheme in vehicular networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 6, pp. 6147–6160, Jun. 2024.
- [55] N. Xu et al., "An efficient and multi-dimensional privacy-preserving platoon communication scheme in vehicular networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 5, pp. 6831–6847, May 2025.
- [56] Y. Sun et al., "PPDR: A privacy-preserving dual reputation management scheme in vehicle platoon," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 6, pp. 6636–6652, Nov./Dec. 2025.
- [57] C.-I. Fan, L.-Y. Huang, and P.-H. Ho, "Anonymous multireceiver identity-based encryption," *IEEE Trans. Comput.*, vol. 59, no. 9, pp. 1239–1249, Sep. 2010.
- [58] B. Waters, "Ciphertext-policy attribute-based encryption: An expressive, efficient, and provably secure realization," in *Proc. IACR Int. Conf. Public-Key Cryptogr.*, 2011, pp. 53–70.
- [59] J. Lai, R. H. Deng, and Y. Li, "Expressive CP-ABE with partially hidden access structures," in *Proc. ACM Symp. Inf., Comput. Commun. Secur.*, 2012, pp. 18–19.
- [60] D. M. C. Willemsen, A. J. C. Schmeitz, and E. Mascalchi, "EU ENSEMBLE project: Specification of an interoperable solution for a support function for platooning," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 6, pp. 5705–5718, Jun. 2023.
- [61] A. Kong, C. Xu, X. Wu, X. Chen, and P. Hang, "Towards emergency scenarios: An integrated decision-making framework of multi-lane platoon reorganization," *IEEE Trans. Consum. Electron.*, vol. 71, no. 4, pp. 11719–11733, Nov. 2025.
- [62] Z. Liu, Z. Cao, and D. S. Wong, "White-box traceable ciphertext-policy attribute-based encryption supporting any monotone access structures," *IEEE Trans. Inf. Forensics Security*, vol. 8, no. 1, pp. 76–88, 2013.
- [63] R. Canetti, S. Halevi, and J. Katz, "Chosen-ciphertext security from identity-based encryption," in *Proc. Adv. Cryptology-EUROCRYPT'04*, 2004, pp. 207–222.
- [64] D. Boneh, B. Lynn, and H. Shacham, "Short signatures from the weil pairing," in *Proc. Adv. Cryptology-ASIACRYPT'01*, 2001, pp. 514–532.
- [65] A. Kate and I. Goldberg, "Distributed private-key generators for identity-based cryptography," in *Proc. Int. Conf. Secur. Cryptogr. Netw.*, 2010, pp. 436–453.
- [66] S. S. M. Chow, "Removing escrow from identity-based encryption," in *Proc. Int. Conf. Pract. Theory Public Key Cryptogr.*, 2009, pp. 256–276.
- [67] M. Chase and S. S. Chow, "Improving privacy and security in multi-authority attribute-based encryption," in *Proc. ACM Conf. Comput. Commun. Secur.*, 2009, pp. 121–130.